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Flow control system, method, and apparatus

Abstract

A gas flow control system for delivering a plurality of gas flows. The gas flow control system has a gas flow path extending from a gas inlet to first and second gas outlets. First and second flow restrictors are operably coupled to the gas flow path. First and second valves are operably coupled to the gas flow path such that when both first and second valves are in a fully open state, flows of gas from the first and second gas outlets are split according to the impedances of the first and second flow restrictors.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/468,042, filed Sep. 7, 2021, which is a continuation of U.S. patent application Ser. No. 17/027,333, now U.S. Pat. No. 11,144,075, filed Sep. 21, 2020, which is (1) a continuation in part of U.S. patent application Ser. No. 16/383,844, now U.S. Pat. No. 10,782,710, filed Apr. 15, 2019, which is a continuation of U.S. patent application Ser. No. 15/638,742, now U.S. Pat. No. 10,303,189, filed Jun. 30, 2017, which in turn claims the benefit of U.S. Provisional Patent Application No. 62/357,113, filed Jun. 30, 2016; (2) a continuation in part of U.S. patent application Ser. No. 16/864,117, now U.S. Pat. No. 11,424,148, filed Apr. 30, 2020, which is a continuation of U.S. patent application Ser. No. 15/717,562, now U.S. Pat. No. 10,679,880, filed Sep. 27, 2017, which in turn claims the benefit of U.S. Provisional Patent Application No. 62/400,324, filed Sep. 27, 2016; (3) a continuation in part of U.S. patent application Ser. No. 16/282,737, now U.S. Pat. No. 10,838,437, filed Feb. 22, 2019, which in turn claims the benefit of U.S. Provisional Patent Application No. 62/633,945, filed Feb. 22, 2018; (4) a continuation in part of U.S. patent application Ser. No. 16/985,540, now U.S. Pat. No. 11,639,865, filed Aug. 5, 2020, which in turn claims the benefit of U.S. Provisional Patent Application No. 62/882,794, filed Aug. 5, 2019; and (5) a continuation in part of U.S. patent application Ser. No.

16/985,635, now U.S. Pat. No. 11,585,444, filed Aug. 5, 2020, which in turn claims the benefit of U.S. Provisional Patent Application No. 62/882,814, filed Aug. 5, 2019, the entireties of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) Mass flow control has been one of the key technologies in semiconductor chip fabrication. Mass flow controllers (MFCs) are important components for delivering process gases for semiconductor fabrication. An MFC is a device used to measure and control the flow of fluids and gases.

(2) As the technology of chip fabrication has improved, so has the demand on the MFC. Semiconductor fabrication processing increasingly requires increased performance, including more accurate measurements, lower equipment costs, greater speed, more consistency in timing in the delivery of gases, and space-saving layouts.

SUMMARY OF THE INVENTION

(3) The present technology is directed to a method of making a plurality of mass flow controllers having different operating characteristics while maintaining a common monolithic base. Thus, different flow components such as cap components, laminar flow components, control valve components, pressure transducer components, volumetric expander components, on/off valve components, and temperature sensor components may be coupled to substantially identical monolithic bases.

(4) The present technology is further directed to a method of improving the transient turn on performance of a pressure based apparatus for controlling flow. This is achieved by pre-pressurizing a volume within the apparatus prior to opening a valve, the valve allowing gas to flow out an outlet of the apparatus.

(5) The present technology is further directed to an apparatus for splitting a flow of process gases into two separate mass flows for use at different locations in a processing chamber.

(6) The present technology is further directed to a laminar flow restrictor for use in a mass flow controller or other gas delivery device. One or more of these gas delivery devices may be used in a wide range of processes such as semiconductor chip fabrication, solar panel fabrication, etc.

(7) The present technology is directed to a seal for a flow restrictor for use in a mass flow controller or other gas delivery device. One or more of these gas delivery devices may be used in a wide range of processes such as semiconductor chip fabrication, solar panel fabrication, etc.

(8) In one implementation, the method of making mass flow controllers comprises providing a plurality of substantially identical monolithic bases, each of the monolithic bases comprising a gas inlet, a gas outlet, and a plurality of flow component mounting regions. The method further comprises coupling a first set of flow components to the flow component mounting regions of a first one of the monolithic bases so that a fluid path is formed from the gas inlet to the gas outlet of the first one of the monolithic bases to which each component of the first set of flow components is in fluid communication, thereby creating a first mass flow controller having a first set of operating characteristics. Finally, a second set of flow components are coupled to the flow component mounting regions of a second one of the monolithic bases so that a fluid path is formed from the gas inlet to the gas outlet of the second one of the monolithic bases to which each component of the second set of flow components is in fluid communication, thereby creating a second mass flow controller having a second set of operating characteristics that are different than the first set of operating characteristics.

(9) In another implementation, the method of making mass flow controllers having different operating characteristics comprises providing a plurality of substantially identical monolithic bases, each of the monolithic bases comprising a gas inlet, a gas outlet, and a plurality of flow component mounting regions. The method further comprises coupling a first set of flow components to the flow component mounting regions of a first one of the monolithic bases, thereby creating a first

mass flow controller having a first set of operating characteristics. Finally, a second set of flow components are coupled to the flow component mounting regions of a second one of the monolithic bases, thereby creating a second mass flow controller having a second set of operating characteristics that are different than the first set of operating characteristics.

(10) In yet another implementation, the method comprises providing a plurality of substantially identical monolithic bases, each of the monolithic bases comprising a gas inlet, a gas outlet, and a plurality of flow component mounting regions. A first set of flow components are coupled to the flow component mounting regions of a first one of the monolithic bases. Lastly, a second set of flow components are coupled to the flow component mounting regions of a first one of the monolithic bases, wherein the first and second sets of flow components comprise different types of flow components.

(11) In another implementation, the invention is a mass flow control apparatus having a monolithic base. The monolithic base has a gas inlet, a gas outlet, a first flow component mounting region, a second flow component mounting region, and a third flow component mounting region. The first flow component mounting region has a first inlet port and a first outlet port, the first inlet port being fluidly coupled to the gas inlet of the monolithic base. The second flow component mounting region has a second inlet port, a second outlet port, and a first auxiliary port.

(12) In yet another implementation, the invention is a mass flow control apparatus having a monolithic base. The monolithic base has a gas inlet, a gas outlet, a first flow component mounting region, a second flow component mounting region, and a third flow component mounting region. A first valve is coupled to the first flow component mounting region. A cap component is coupled to one of the second or third flow component mounting regions. The first flow component region has a first inlet port and a first outlet port, the first inlet port being fluidly coupled to the gas inlet of the monolithic base. The second flow component mounting region comprises a second inlet port, a second outlet port, and a first auxiliary port.

(13) In a further implementation, the invention is a mass flow control apparatus having a monolithic base. The monolithic base has a gas inlet, a gas outlet, a first flow component mounting region, a second flow component mounting region, and a third flow component mounting region. A first valve is coupled to the first flow component mounting region. A second valve is coupled to the second flow component mounting region. A third valve is coupled to the third flow component mounting region. The first flow component mounting region has a first inlet port fluidly coupled to the gas inlet of the monolithic base. The second flow component mounting region has a first auxiliary port. The third valve is fluidly coupled to the gas outlet of the monolithic base.

(14) In one embodiment, the invention is a method of delivering a gas at a predetermined flow rate. A gas flow control apparatus is provided, the gas flow control apparatus having a gas flow path extending from a gas inlet to a gas outlet, a proportional valve coupled to the flow path, an on/off valve coupled to the flow path and downstream of the proportional valve, a volume of the gas flow path defined between the proportional valve and the on/off valve. A flow restrictor having a flow impedance is located downstream of the proportional valve. The volume of the apparatus is pressurized with the gas to a target pre-flow pressure by opening the proportional valve while the on/off valve is off, the target pre-flow pressure being selected to achieve the predetermined flow rate. Finally, the on/off valve is opened to deliver the gas to the gas outlet at the predetermined flow rate.

(15) In another embodiment, the invention is a system for delivering a gas at a predetermined flow rate. The system has a gas flow control apparatus having a gas flow path extending from a gas inlet to a gas outlet, a proportional valve coupled to the flow path, an on/off valve coupled to the flow path and downstream of the proportional valve, a volume of the gas flow path defined between the proportional valve and the on/off valve. A flow restrictor having a flow impedance is located downstream of the proportional valve. A controller pressurizes the volume with the gas to a target pre-flow pressure by opening the proportional valve while the on/off valve is off, the target pre-

flow pressure being selected to achieve the predetermined flow rate. Finally, the controller opens the on/off valve to deliver the gas to the gas outlet at the predetermined flow rate.

(16) In yet another embodiment, the invention is a method of delivering gas at a predetermined flow rate. A controller generates a gas flow activation signal at a first time, the signal comprising data identifying a second time at which the gas is to be delivered from a gas outlet of a gas flow path at the predetermined flow rate. The first time is prior to the second time and a priming period is defined as the time between the first and second times. Upon receipt of the gas flow activation signal, the controller adjusts one or more components of a gas flow apparatus to achieve a primed condition of the gas in a volume of the gas flow path during the priming period, the priming period selected to achieve the predetermined flow rate. During the priming period, the gas is prohibited from exiting the gas outlet of the gas flow path. Finally, gas is delivered from the volume at the second time via the gas outlet of the gas flow path.

(17) In another embodiment, the invention is a system for delivering a gas at a predetermined flow rate, the system having a gas flow path extending from a gas inlet to a gas outlet, one or more components configured to define a volume of the gas flow path and control flow of the gas through the gas flow path, and a controller. The controller is configured to generate a gas flow activation signal at a first time, the signal identifying a second time at which the gas is to be delivered from the gas outlet, the first time being prior to the second time and a priming period being defined between the first and second times. The controller is also configured to adjust the one or more components to achieve a primed condition of the gas in the volume during the priming period upon receipt of the gas flow activation signal, the primed condition selected to achieve the predetermined flow rate. The gas is prohibited from exiting the gas outlet of the gas flow path during the priming period. Finally, the controller is configured to adjust the one or more components to deliver the gas from the volume at the second time.

(18) In a further embodiment, the invention is a method of delivering gas at a predetermined flow rate. The method involves priming, in a volume of a gas flow path and during a priming period, a gas to a primed condition, the primed condition selected to achieve the predetermined flow rate. The gas is prohibited from exiting a gas outlet of the gas flow path during the priming period. Second, gas is delivered from the volume subsequent to the priming period.

(19) In one embodiment, the invention is a gas flow control system for delivering a plurality of gas flows. The gas flow control system includes a gas flow path extending from a gas inlet to a first gas outlet and a second gas outlet. A first on/off valve is operably coupled to the gas flow path and is located between the gas inlet and the first gas outlet. A second on/off valve is operably coupled to the gas flow path and located between the gas inlet and the second gas outlet. A first flow restrictor having a first flow impedance is operably coupled to the gas flow path and located between the gas inlet and the first gas outlet. A second flow restrictor having a second flow impedance is operably coupled to the gas flow path and located between the gas inlet and the second gas outlet. When both the first and second on/off valves are in a fully open state, a ratio between a first gas flow from the first gas outlet and a second gas flow from the second gas outlet is determined by a ratio of the first flow impedance and the second flow impedance.

(20) In another embodiment, the invention is a gas flow control system for delivering a plurality of gas flows including a gas flow path extending from a gas inlet to a first gas outlet and a second gas outlet. A first valve is operably coupled to the gas flow path and located between the gas inlet and the first gas outlet. A second valve is operably coupled to the gas flow path and located between the gas inlet and the second gas outlet. A first flow restrictor having a first flow impedance is operably coupled to the gas flow path and located between the gas inlet and the first gas outlet. A second flow restrictor having a second flow impedance is operably coupled to the gas flow path and located between the gas inlet and the second gas outlet.

(21) In yet another embodiment, the invention is a method of delivering a process gas. The method includes providing a gas flow apparatus having a gas flow path extending from a gas inlet to a first

gas outlet and a second gas outlet, a first valve operably coupled to the gas flow path and located between the gas inlet and the first gas outlet, a second valve operably coupled to the gas flow path and located between the gas inlet and the second gas outlet, a first flow restrictor having a first flow impedance operably coupled to the gas flow path and located between the gas inlet and the first gas outlet, and a second flow restrictor having a second flow impedance operably coupled to the gas flow path and located between the gas inlet and the second gas outlet. The first valve is transitioned to a fully open state and the second valve to a fully closed state to deliver a first controlled flow of process gas to the first gas outlet. Subsequently, the second valve is transitioned to a fully open state and the first valve to a fully closed state to deliver a second controlled flow of process gas to the second gas outlet. Finally, the first valve is transitioned to a fully open state to deliver the first controlled flow of process gas to the first gas outlet simultaneously with delivering the second controlled flow of process gas to the second gas outlet.

(22) In one implementation, the invention is a flow restrictor for restricting the flow of a gas. The flow restrictor has a first end, a second end, and a longitudinal axis extending from the first end to the second end. A plurality of first layers extend from the first end to the second end along the longitudinal axis. A plurality of second layers extend from the first end to the second end along the longitudinal axis. A first aperture at the first end is defined by the plurality of first layers and the plurality of second layers. A second aperture at the second end is defined by the plurality of first layers and the plurality of second layers. A flow passage is defined by the plurality of first layers and the plurality of second layers, the flow passage extending from the first aperture to the second aperture.

(23) In another implementation, the invention is a mass flow control apparatus for delivery of a fluid, the mass flow control apparatus having a valve comprising an inlet passage, an outlet passage, a valve seat, and a closure member. The mass flow control apparatus also has a flow restrictor, the flow restrictor positioned in one of the inlet passage or the outlet passage. The flow restrictor has a first end, a second end, and a longitudinal axis extending from the first end to the second end. A plurality of layers extend substantially parallel to the longitudinal axis. A first aperture is located at the first end and a second aperture is located at the second end. A flow passage is defined by the plurality of layers, the flow passage fluidly coupled to the first aperture and the second aperture.

(24) In yet another implementation, the invention is a method of manufacturing a flow restrictor. First, a plurality of layer blanks are provided, the layer blanks having a first edge, a second edge opposite the first edge, a third edge, a fourth edge opposite the third edge, a front face, and a rear face opposite the front face. A first cavity is formed in the front face of a first one of the plurality of layer blanks. The plurality of layer blanks are stacked. Subsequently, the plurality of layer blanks are bonded to form a resistor stack having a first unfinished end and an opposite second unfinished end. The first unfinished end of the resistor stack is formed by the first edges of the plurality of layer blanks and the second unfinished end of the resistor stack is formed by the second edges of the plurality of layer blanks. Finally, material is removed from the first unfinished end of the layer stack to expose the first cavity and form a first aperture.

(25) In one implementation, the invention is a seal for a gas flow restrictor, the seal having a first end, a second end, and an aperture for receiving the flow restrictor to form a fluid tight connection between the flow restrictor and the seal.

(26) In another implementation, the invention is a valve assembly, the valve assembly having a valve, a flow restrictor, and a seal. The valve has a passage. The flow restrictor has a first end, a second end, a longitudinal axis extending from the first end to the second end, and a sealing portion located between the first end and the second end along the longitudinal axis. The seal is in contact with the sealing portion of the flow restrictor and the passage of the valve.

(27) In yet a further implementation, the invention is a valve assembly, the valve assembly having a valve, the valve having a first passage, a second passage, a first sealing recess, and a second recess.

The valve assembly has a base having a third sealing recess and a fourth sealing recess. The valve assembly has a flow restrictor, the flow restrictor having a first end, a second end, a longitudinal axis extending from the first end to the second end, and a surface of the flow restrictor located between the first end and the second end along the longitudinal axis. Finally, the valve assembly has a seal in contact with the surface of the flow restrictor and the first sealing recess of the valve.

(28) In yet another embodiment, the invention is a mass flow control apparatus having a monolithic base, the monolithic base having a gas inlet, a gas outlet, a first flow component mounting region, a second flow component mounting region, and a third flow component mounting region. The first flow component mounting region has a first inlet port and a first outlet port, the first inlet port fluidly coupled to the gas inlet of the monolithic base. The third flow component mounting region has a first sensing port fluidly coupled to the gas outlet of the monolithic base.

(29) In a further embodiment, the invention is a mass flow control apparatus having a monolithic base, the monolithic base comprising a gas inlet, a gas outlet, a first flow component mounting region, a second flow component mounting region, and a third flow component mounting region. A first valve is coupled to the first flow component mounting region. A cap component is coupled to the second flow component mounting region. A second valve is fluidly coupled to the third flow component mounting region. The first flow component mounting region comprises a first inlet port and a first outlet port, the first inlet port fluidly coupled to the gas inlet of the monolithic base.

(30) In another embodiment, the invention is a gas flow control system for delivering a gas at a predetermined flow rate, the gas flow control system having a gas flow path extending from a gas inlet to a gas outlet. A proportional valve is operably coupled to the gas flow path. An on/off valve is operably coupled to the gas flow path downstream of the proportional valve, a volume of the gas flow path being defined between the proportional valve and the on/off valve. A flow restrictor having a flow impedance is located downstream of the proportional valve. A bleed valve is operably coupled to the volume of the gas flow path. Finally, a controller is configured to 1) pressurize the volume with the gas to a target pre-flow pressure by opening the proportional valve while the on/off valve is in an off-state, the target pre-flow pressure selected to achieve a predetermined flow rate, 2) open the bleed valve by moving the bleed valve to an on-state and flow the gas at the predetermined flow rate through the bleed valve, and 3) simultaneously open the on/off valve by moving the on/off valve to an on-state and close the bleed valve by moving the bleed valve to an off-state to deliver the gas to the gas outlet at the predetermined flow rate.

(31) In yet another embodiment, the invention is a flow restrictor for restricting the flow of a gas, the flow restrictor having a first end, a second end, and a longitudinal axis extending from the first end to the second end. A plurality of layers extend from the first end to the second end along the longitudinal axis. A first aperture is located at the first end and is defined by the plurality of layers. A second aperture is located at the second end and is defined by the plurality of layers. A flow passage is defined by the plurality of layers, the flow passage extending from the first aperture to the second aperture.

(32) In a further embodiment, the invention is a flow restrictor assembly including a flow restrictor and a seal. The flow restrictor has a first end, a second end, a longitudinal axis extending from the first end to the second end, and a sealing portion located between the first end and the second end along the longitudinal axis. The seal is in contact with the sealing portion of the flow restrictor.

(33) In another embodiment, the invention is a gas flow control system for delivering a plurality of gas flows, the system including a gas flow path extending from a gas inlet to a first gas outlet and a second gas outlet. The system further includes a first proportional valve operably coupled to the gas flow path and a first on/off valve operably coupled to the gas flow path and located between the first proportional valve and the first gas outlet. Furthermore, the system includes a first flow restrictor having a first flow impedance operably coupled to the gas flow path and located between the first proportional valve and the first gas outlet and a second flow restrictor having a second flow impedance operably coupled to the gas flow path and located between the first proportional

valve and the second gas outlet. When both the first proportional valve and the first on/off valve are in a fully open state, a ratio between a first gas flow from the first gas outlet and a second gas flow from the second gas outlet is determined by a ratio of the first flow impedance and the second flow impedance.

(34) In yet another embodiment, the invention is a gas flow control system for delivering a plurality of gas flows, the system including a gas flow path extending from a gas inlet to a first gas outlet and a second gas outlet. A first valve is operably coupled to the gas flow path and located between the gas inlet and the first gas outlet. A second valve is operably coupled to the gas flow path and located between the gas inlet and the second gas outlet. A first flow restrictor having a first flow impedance is operably coupled to the gas flow path. A second flow restrictor having a second flow impedance is operably coupled to the gas flow path. When both the first valve and the second valve are in a fully open state, a ratio between a first gas flow from the first gas outlet and a second gas flow from the second gas outlet is determined by a ratio of the first flow impedance and the second flow impedance.

(35) In another embodiment, the invention is a method of delivering a process gas. In a first step, a gas flow apparatus is provided, the gas flow apparatus having a gas flow path extending from a gas inlet to a first gas outlet and a second gas outlet, a first valve operably coupled to the gas flow path and located between the gas inlet and the first gas outlet, a second valve operably coupled to the gas flow path and located between the gas inlet and the second gas outlet, a first flow restrictor having a first flow impedance operably coupled to the gas flow path and located between the gas inlet and the first gas outlet, and a second flow restrictor having a second flow impedance operably coupled to the gas flow path and located between the gas inlet and the second gas outlet. In a second step, the first valve is transitioned to a partially open state and a first controlled flow of process gas is delivered to the first gas outlet. In a third step, the second valve is transitioned to a fully open state and a second controlled flow of process gas is delivered to the second gas outlet.

(36) In yet another embodiment, the invention is a method of manufacturing a semiconductor device. In a first step, a gas is supplied to a gas inlet of a gas flow control apparatus. The gas flow control apparatus has a gas flow path extending from the gas inlet to a gas outlet; a proportional valve operably coupled to the gas flow path; an on/off valve operably coupled to the gas flow path between the proportional valve and the gas outlet, a volume of the gas flow path being defined between the proportional valve and the on/off valve; and a flow restrictor having a flow impedance operably coupled to the gas flow path and located between the proportional valve and the gas outlet. In a second step, the volume is pressurized with the gas to a target pre-flow pressure by opening the proportional valve while the on/off valve is in an off-state, the target pre-flow pressure selected, in view of the flow impedance of the flow restrictor, to achieve the predetermined flow rate. In a third step, the on/off valve is opened by moving the on/off valve to an on-state, thereby delivering the gas to the gas outlet at the predetermined flow rate. In a fourth step, the gas is flowed from the gas outlet to a semiconductor manufacturing process.

(37) In another embodiment, the invention is a method of manufacturing a semiconductor device. In a first step, a gas is supplied to a gas inlet of a gas flow control apparatus. In a second step, a controller generates a gas flow activation signal at a first time that includes data identifying a second time at which the gas is to be delivered from a gas outlet of a gas flow path of the gas flow control apparatus at a predetermined flow rate, the first time being prior to the second time and a priming period being defined between the first and second times. In a third step, the controller receives the gas flow activation signal and, in response, adjusts one or more components of the gas flow control apparatus to achieve a primed condition of the gas in a volume of the gas flow path during the priming period, the primed condition selected to achieve the predetermined flow rate, the gas being prohibited from exiting the gas outlet of the gas flow path during the priming period. In a fourth step, the gas is delivered from the volume at the second time via the gas outlet of the gas flow path to a semiconductor manufacturing process.

(38) In yet another embodiment, the invention is a method of manufacturing a semiconductor device. In a first step, a gas is supplied to a gas inlet of a gas flow control apparatus. In a second step, a volume of a gas flow path is primed during a priming period with the gas to a primed condition, the primed condition selected to achieve the predetermined flow rate, the gas being prohibited from exiting a gas outlet of the gas flow path during the priming period. In a third step, the gas is delivered from the volume of the gas flow control apparatus to a semiconductor manufacturing process subsequent to the priming period.

(39) Further areas of applicability of the present technology will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred implementation, are intended for purposes of illustration only and are not intended to limit the scope of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention of the present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. 1 is a perspective view of a first embodiment of the monolithic base.

(3) FIG. 2 is a lower perspective view of the monolithic base of FIG. 1.

(4) FIG. 3 is a top view of the monolithic base of FIG. 1.

(5) FIG. 4 is a bottom view of the monolithic base of FIG. 1.

(6) FIG. 5 is a cross sectional view taken along the line V-V of FIG. 4.

(7) FIG. 6 is a perspective view of a second embodiment of the monolithic base.

(8) FIG. 7 is a lower perspective view of the monolithic base of FIG. 6.

(9) FIG. 8 is a top view of the monolithic base of FIG. 6.

(10) FIG. 9 is a bottom view of the monolithic base of FIG. 6.

(11) FIG. 10 is a cross sectional view taken along the line X-X of FIG. 8.

(12) FIG. 11 is a side view of a mass flow controller in accordance with a first embodiment.

(13) FIG. 12 is a cross sectional view of the mass flow controller of FIG. 11 illustrating the ports and flow passages.

(14) FIG. 13 is a side view of a mass flow controller in accordance with a second embodiment.

(15) FIG. 14 is a cross sectional view of the mass flow controller of FIG. 13 illustrating the ports and flow passages.

(16) FIG. 15 is a side view of a mass flow controller in accordance with a third embodiment.

(17) FIG. 16 is a cross sectional view of the mass flow controller of FIG. 15 illustrating the ports and flow passages.

(18) FIG. 17 is a side view of a mass flow controller in accordance with a fourth embodiment.

(19) FIG. 18 is a cross sectional view of the mass flow controller of FIG. 17 illustrating the ports and flow passages.

(20) FIG. 19 is a side view of a mass flow controller in accordance with a fifth embodiment.

(21) FIG. 20 is a cross sectional view of the mass flow controller of FIG. 19 illustrating the ports and flow passages.

(22) FIG. 21 is a side view of a mass flow controller in accordance with a sixth embodiment.

(23) FIG. 22 is a cross sectional view of the mass flow controller of FIG. 21 illustrating the ports and flow passages.

(24) FIG. 23 is a side view of a mass flow controller in accordance with a seventh embodiment.

(25) FIG. 24 is a cross sectional view of the mass flow controller of FIG. 23 illustrating the ports and flow passages.

(26) FIG. 25 is a cross sectional view of an apparatus for controlling gas flow.

- (27) FIG. 26 is a block diagram illustrating the control system for the apparatus of FIG. 25.
- (28) FIG. 27 is a schematic diagram of the apparatus of FIG. 25.
- (29) FIG. 28 is a flow chart illustrating a first method of improving transient turn on performance.
- (30) FIG. 29 is a graph illustrating normalized mass flow rates resulting from implementation of the method of FIG. 28.
- (31) FIG. 30A is a cross sectional view of another embodiment of an apparatus for controlling gas flow.
- (32) FIG. 30B is a cross sectional view of another embodiment of an apparatus for controlling gas flow.
- (33) FIG. 31 is a flow chart illustrating a second method of improving transient turn on performance.
- (34) FIG. 32 is a flow chart illustrating a third method of improving transient turn on performance.
- (35) FIG. 33 is a graph illustrating normalized mass flow rates resulting from implementation of the method of FIG. 32.
- (36) FIG. 34 is a flow chart illustrating a fourth method of improving transient turn on performance.
- (37) FIG. 35 is a schematic view of an apparatus for controlling gas flow showing the flow path of the process gases through the apparatus.
- (38) FIG. 36 is a block diagram illustrating the control system for the apparatus of FIG. 35.
- (39) FIG. 37 is a schematic diagram of the apparatus of FIG. 35.
- (40) FIG. 38 is a schematic diagram of another embodiment of the invention having an additional proportional valve.
- (41) FIG. 39 is a schematic diagram of another embodiment of the invention having three gas outlets.
- (42) FIG. 40 is a schematic diagram of another embodiment of the invention utilizing two proportional valves and a single on/off valve.
- (43) FIG. 41 is a schematic of a process utilizing one or more laminar flow restrictors.
- (44) FIG. 42 is a schematic of a mass flow controller as may be utilized in the process of FIG. 41.
- (45) FIG. 43 is a perspective view of a first embodiment of a laminar flow restrictor as may be utilized in the mass flow controller of FIG. 42.
- (46) FIG. 44 is a perspective view illustrating a portion of the layers forming the flow restrictor of FIG. 43.
- (47) FIG. 45A is an end view of the portion of the flow restrictor of FIG. 44.
- (48) FIG. 45B is a detail view of the area XLVB of FIG. 45A.
- (49) FIG. 46 is an exploded perspective view of the portion of the flow restrictor of FIG. 44.
- (50) FIG. 47 is a cross-sectional view of the portion of the flow restrictor of FIG. 44, taken along line XLVII-XLVII.
- (51) FIG. 48 is a top view of a first layer of the flow restrictor of FIG. 43.
- (52) FIG. 49 is a top view of a second layer of the flow restrictor of FIG. 43.
- (53) FIG. 50 is a perspective view of a second embodiment of a laminar flow restrictor.
- (54) FIG. 51 is a perspective view illustrating a portion of the layers forming the flow restrictor of FIG. 50.
- (55) FIG. 52A is an end view of the portion of the flow restrictor of FIG. 51.
- (56) FIG. 52B is a detail view of the area LIIB of FIG. 52A.
- (57) FIG. 53 is an exploded perspective view of the portion of the flow restrictor of FIG. 51.
- (58) FIG. 54 is a cross-sectional view of the portion of the flow restrictor of FIG. 51, taken along line LIV-LIV.
- (59) FIG. 55 is a top view a first layer of the flow restrictor of FIG. 50.
- (60) FIG. 56 is a top view a second layer of the flow restrictor of FIG. 50.
- (61) FIG. 57 is a perspective view of a portion of a third embodiment of a laminar flow restrictor.

(62) FIG. 58 is an end view of the portion of the flow restrictor of FIG. 57.

(63) FIG. 59 is an exploded perspective view of the portion of the flow restrictor of FIG. 57.

(64) FIG. 60 is a cross-sectional view of the portion of the flow restrictor of FIG. 57, taken along line LX-LX.

(65) FIG. 61 is a top view of a first layer of the flow restrictor of FIG. 57.

(66) FIG. 62 is a top view of a second layer of the flow restrictor of FIG. 57.

(67) FIG. 63 is a perspective view of a portion of a fourth embodiment of a laminar flow restrictor.

(68) FIG. 64 is an end view of the portion of the flow restrictor of FIG. 63.

(69) FIG. 65 is an exploded perspective view of the portion of the flow restrictor of FIG. 63.

(70) FIG. 66 is a cross-sectional view of the portion of the flow restrictor of FIG. 63, taken along line LXVI-LXVI.

(71) FIG. 67 is a top view of a first layer of the flow restrictor of FIG. 63.

(72) FIG. 68 is a top view of a second layer of the flow restrictor of FIG. 63.

(73) FIG. 69 is a top view of a third layer of the flow restrictor of FIG. 63.

(74) FIG. 70 is a perspective view of a fifth embodiment of a laminar flow restrictor.

(75) FIG. 71 is a perspective view illustrating a portion of the layers forming the flow restrictor of FIG. 70.

(76) FIG. 72 is an end view of the portion of the flow restrictor of FIG. 71.

(77) FIG. 73 is an exploded perspective view of the portion of the flow restrictor of FIG. 71.

(78) FIG. 74 is a cross-sectional view of the portion of the flow restrictor of FIG. 71, taken along line LXXIV-LXXIV.

(79) FIG. 75 is a top view a first layer of the flow restrictor of FIG. 71.

(80) FIG. 76 is a top view a second layer of the flow restrictor of FIG. 71.

(81) FIG. 77 is an exploded perspective view of a plurality of layer blanks illustrating methods of manufacturing the disclosed flow restrictors.

(82) FIG. 78 is a top view of a first layer of the invention of FIG. 77.

(83) FIG. 79 is a top view of a second layer of the invention of FIG. 77.

(84) FIG. 80 is a perspective view of a restrictor stack prior to finishing according to the invention of FIG. 77.

(85) FIG. 81 is a perspective view of a restrictor stack after finishing according to the invention of FIG. 77.

(86) FIG. 82 is a schematic of a process utilizing one or more flow restrictors.

(87) FIG. 83 is a schematic of a mass flow controller as may be utilized in the process of FIG. 82.

(88) FIG. 84 is a schematic view of a valve incorporating a first embodiment of a flow restrictor and seal as may be utilized in the mass flow controller of FIG. 83.

(89) FIG. 85 is a perspective view of the first embodiment of the flow restrictor and seal as may be utilized in the valve of FIG. 84.

(90) FIG. 86 is a cross-sectional view of the flow restrictor and seal of FIG. 85, taken along line LXXXV-LXXXV.

(91) FIG. 87 is a perspective view of the flow restrictor of FIG. 85 without the seal.

(92) FIG. 88 is a perspective view of the seal of FIG. 85 without the flow restrictor.

(93) FIG. 89 is a schematic view of a valve incorporating a second embodiment of a flow restrictor and seal as may be utilized in the mass flow controller of FIG. 83.

(94) FIG. 90 is a perspective view of the second embodiment of the flow restrictor and seal as may be utilized in the valve of FIG. 89.

(95) FIG. 91 is a cross-sectional view of the flow restrictor and seal of FIG. 90, taken along line XCI-XCI.

(96) FIG. 92 is a perspective view of the seal of FIG. 90 without the flow restrictor.

(97) FIG. 93 is a cross-sectional view of the seal of FIG. 92, taken along line XCIII-XCIII.

(98) FIG. 94 is a front view of the seal of FIG. 92.

(99) FIG. 95 is a top view of the seal of FIG. 92.

DETAILED DESCRIPTION

(100) The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “left,” “right,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the preferred embodiments. Accordingly, the invention expressly should not be limited to such preferred embodiments illustrating some possible non-limiting combinations of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

(101) The disclosure is divided into four sections. Section I discusses a modular method of making mass flow controllers. Section II discusses a method of improving the transient turn on performance of pressure based apparatuses for controlling mass flow. Section III discusses an apparatus for splitting a flow of process gas into two individual flows of process gas having a known ratio. Section IV discusses laminar flow restrictors for use in an apparatus for controlling gas flow. Section V discusses seals for laminar flow restrictors for use in an apparatus for controlling gas flow. Different embodiments disclosed in the respective sections can be used together as part of a gas delivery apparatus, method, or system. To the extent a term, reference number, or symbol is used differently in different sections, context should be taken from the relevant section and not the other sections.

Section I

(102) The present invention is directed to a modular method of making mass flow controllers which reduces total system cost and time to production. Customers require a variety of mass flow controllers within a single semiconductor process to permit them to apply a diverse array of gases in a wide range of mass flow rates. In an effort to accommodate this need, manufacturers of mass flow control equipment have designed mass flow controllers with a variety of component configurations to achieve specific operating characteristics. This often requires different component configurations. Due to the limited space available for mass flow controllers, a custom configuration for the mounting base is typically required. This is often accomplished through a plurality of mounting base components which are assembled to meet particular needs. Alternately, a custom mounting base may be fabricated which is unitary and incorporates all of the necessary ports and passages required for the mass flow controller currently being manufactured. However, due to the wide variation in operating characteristics required in a product line, the mounting bases are customized for the specific application and desired operating characteristics.

(103) FIG. 1 shows a perspective view of a first embodiment of a monolithic base **100** according to the present invention. FIGS. 2-5 show the monolithic base in greater detail. The monolithic base **100** has a plurality of flow component mounting regions where flow components may be mounted. The plurality of flow component mounting regions include a first flow component mounting region **110**, a second flow component mounting region **120**, and a third flow component mounting region **130**. In the present embodiment of the monolithic base **100**, the first flow component mounting

region **110** and the second flow component mounting region **120** are both capable of being used as multi-function ports which may incorporate a variety of flow components. In contrast, the third flow component mounting region **130** is a sensing port intended to accept a pressure transducer component.

(104) Turning to FIG. 5, a cross section of the monolithic base **100** is shown. A gas inlet **150** is located on the lower left side of the monolithic base **100**. When the completed mass flow controller is installed in process machinery, process gas is supplied to the gas inlet **150**. It then flows downstream to the first flow component mounting region **110**. The first flow component mounting region **110** has an inlet port **112** that is directly connected to the gas inlet **150** by a first flow passage **152**. The first flow component region **110** also has an outlet port **114** that is not fluidly connected to the inlet port **112**.

(105) The outlet port **114** of the first flow component mounting region **110** is fluidly connected to an inlet port **122** of the second flow component mounting region **120** by a second flow passage **154**. The second flow component mounting region **120** also contains an outlet port **124** and an auxiliary port **126**. The auxiliary port **126** feeds an auxiliary passage **156** which has a gas vent **157** located on the underside of the monolithic base **100**. The outlet port **124** of the second flow component mounting region **120** is fluidly connected by a third flow passage **158** to a gas outlet **160** located underneath the right side of the monolithic base **100**. The third flow component mounting region **130** also has a sensing port **132** that is also connected to the third flow passage **158** so that a device attached to the third flow component mounting region **130** can sense the pressure within the third flow passage **158**.

(106) FIGS. 6-10 show a second embodiment of the monolithic base **200**. This embodiment is formed with a greater thickness so that there is adequate room to provide for additional flow passages and a greater number of flow component mounting regions. As best seen in FIG. 6, the monolithic base **200** has a first flow component mounting region **210**, a second flow component mounting region **220**, a third flow component mounting region **230**, and a fourth flow component mounting region **240**. Turning to FIG. 10, a cross section of the monolithic base **200** is provided to better show the internal configuration.

(107) The first flow component mounting region **210** has an inlet port **212** and an outlet port **214**. The second flow component mounting region **220** has an inlet port **222**, an outlet port **224**, and an auxiliary port **226**. The third flow component mounting region **230** has an inlet port **232**, an outlet port **234**, and an auxiliary port **236**. The fourth flow component mounting region **240** has an inlet port **242** and an outlet port **244**.

(108) Similar to the monolithic base **100** discussed above, process gas is supplied to a mass flow controller at the gas inlet **250** located on the underside of the monolithic base **200**. The process gas then flows through a first flow passage **252** to the inlet port **212** of the first flow component mounting region. The outlet port **214** of the first flow component mounting region **210** is connected by a second flow passage **254** to the inlet port **222** of the second flow component mounting region **220**. The auxiliary port **226** of the second flow component mounting region **220** also connects to an auxiliary passage **256** which rims between the auxiliary port **226** and a gas vent **257** located on the underside of the monolithic base **200**. The outlet port **224** of the second flow component mounting region **220** is connected to the inlet port **232** of the third flow component mounting region **230** by a third flow passage **258**.

(109) However, unlike the second flow component mounting region **220**, the auxiliary port **236** of the third flow component mounting region **230** is located to the right of the outlet port **234**. The outlet port **234** of the third flow component mounting region **230** is connected to a gas outlet **260** located on the underside of the monolithic base **200** by the fourth flow passage **262**. The fourth flow passage **262** also connects the outlet port **244** of the fourth flow component mounting region **240** to the gas outlet **260**. Finally, the auxiliary port **236** of the third flow component mounting region **230** is connected to the inlet port **242** of the fourth flow component mounting region **240** by

a fifth flow passage **264**. An accessory port **266** is located at the rightmost end of the monolithic base **200**, and may be plugged or welded if the design does not require it. Alternately, the accessory port **266** may be used to attach a pressure transducer component or other flow component.

(110) FIGS. **11** and **12** show a first embodiment of a mass flow controller **300** incorporating the monolithic base **100** discussed above. The mass flow controller **300** also comprises a control valve component **405**, a cap component **410**, a pressure transducer component **415**, a laminar flow component **420**, and a substrate block **425**. In this embodiment, the mass flow controller may be sized to provide a desired flow rate by selecting a laminar flow component **420** having an appropriately sized restrictor **421**. The restrictor **421** may be selected so as to change the range of mass flow rates that the mass flow controller **300** may supply. The restrictor **421** may be formed as a porous block, a device having small orifices or channels, or any other means of providing a restriction to process gas flow that is consistent and repeatable across a target dynamic operating range of mass flow rates. The restrictor **421** has a greater resistance to flow than the passages upstream and downstream of the restrictor **421**.

(111) The flow path of the process gas is indicated by arrows which illustrate the path that the process gas takes through the mass flow controller **300**. The process gas provided at the gas inlet **150** is supplied to the inlet port **112** of the first flow component mounting region **110**. The inlet port **112** is fluidly coupled to the control valve component **405**. The control valve component **405** meters the amount of process gas which passes to the outlet port **114**. The control valve component **405** is capable of providing proportional control of the process gas such that it need not be fully open or closed, but instead may have intermediate states to permit control of the mass flow rate of process gas. After the control valve component **405**, process gas passes through the cap component **410**. The cap component **410** which has a passage formed therein to permit unrestricted flow of the gas from the inlet port **122** to the outlet port **124** at the second flow component mounting region **120**. The auxiliary port **126** is blocked off by a blocking seal **430** which prevents process gas from flowing into the auxiliary passage **156**. The cap component **410** also has a port for coupling a pressure transducer component **415** if desired. In the present embodiment, the top port on the cap component **410** is plugged because only one pressure transducer component **415** is required.

(112) Process gas then flows from the outlet port **124** of the second component mounting region **120** through the third flow passage **158** to the gas outlet **160**. The pressure transducer component **415** is coupled to the third flow component mounting region **130**. The pressure transducer **415** samples the pressure of the process gas in the third flow passage **158**. The process gas then flows into the substrate block **425** through a first substrate flow passage **426** to a substrate block inlet port **427**. The laminar flow component **420** is coupled to the substrate block **425**. The substrate block **425** is typically preinstalled on a customer's process equipment, and generally has a standardized arrangement of ports.

(113) Process gas then flows through the laminar flow component **420**, past the restrictor **421**, and through the substrate block outlet port **428**. The laminar flow component **420** incorporates an on/off valve component integrally within the laminar flow component **420** to permit complete shutoff of process gas flow. The substrate gas outlet **429** is connected to a process manifold external to the mass flow controller **300**. The substrate block **425** further comprises an accessory port **431** which may be plugged or welded if not required. Alternately, another pressure transducer component **415** may be attached to the accessory port **431**.

(114) Turning to FIGS. **13** and **14**, a second embodiment of a mass flow controller **500** is shown. This mass flow controller is optimized for extremely low mass flow rates, and comprises a monolithic base **100**, a control valve component **405**, a bleed component **435**, a pressure transducer component **415**, a laminar flow component **420**, and a substrate block **425**. The control valve component **405**, pressure transducer component **415**, the laminar flow component **420**, and the substrate block **425** are equivalent to those disclosed in the mass flow controller **300**.

(115) The bleed component **435** is mounted in the second component mounting region **120** and is

operatively connected to the inlet port **122**, outlet port **124**, and auxiliary port **126**. The bleed component **435** incorporates an orifice **436** which is sized to permit a desired amount of process gas to pass through the orifice **436** and the auxiliary port **126**. The bleed component **435** also incorporates an on/off valve component to enable selective activation of the bleed functionality. The orifice **436** may be formed as one or more holes, a porous element, or any other means of providing a calibrated restriction to gas flow. When the bleed component **435** is activated, process gas is bled through the orifice **436** and passes into the auxiliary passage **156** to the bleed conduit **437**. The bleed conduit **437** is directed to a process vacuum system for recovering and disposing of process gases. The bleed component **435** allows the mass flow controller **500** to have greatly increased accuracy and control at low mass flow rates where the control valve component **405** is unable to effectively control flow. Instead of exclusively relying on the control valve component **405** to meter process gas, the bleed component **435** allows additional control over the process gas which passes through the mass flow controller **500**.

(116) The mass flow controller **600** of FIGS. **15** and **16** is designed to provide an exceptionally stable flow of process gas. This is accomplished by combining a monolithic base **100** with a control valve component **405**, a volumetric expander component **440**, a pressure transducer **415**, a laminar flow component **420**, and a substrate block **425**. As with the mass flow controllers **300**, **500** discussed above, the mass flow controller **600** incorporates the control valve component **405**, pressure transducer **415**, laminar flow component **420**, and substrate block **425** and these components serve the same purpose. However, in place of the bleed component **435**, the volumetric expander component **440** is fitted to provide a known volume to dampen pressure pulses in the process gas. The volumetric expander component **440** increases the stability of the mass flow rate. The auxiliary port **126** of the second flow component mounting region **120** is not connected to the volumetric expander component **440**. Instead, a blocking seal **430** is provided on the auxiliary port **126** to ensure that no gas flow can occur.

(117) FIGS. **17** and **18** illustrate a fourth embodiment of the mass flow controller **700**. The mass flow controller **700** incorporates a monolithic base **200**, a control valve component **405**, a bleed component **435**, a pressure transducer component **415**, a cap component **410**, and a laminar flow component **420**. The mass flow controller **700** is similar to that disclosed in FIGS. **13** and **14**, but differs in that the substrate block **425** has been eliminated and its features incorporated into the monolithic base **200**. The mass flow controller **700** offers the same advantages in low flow accuracy with a different monolithic base. The process gas flows through the control valve component **405** as before, then passes through the bleed component **435** with a portion of the process gas being allowed to bleed through the orifice **436**, through the auxiliary port **226** of the second flow component mounting region **120**, and on through the auxiliary passage **156** to the bleed conduit **437**.

(118) The remaining process gas passes through the third flow passage **258** to the inlet port **232** of the third flow component mounting region **230**. The cap component **410** is coupled to the third flow component mounting region **230** and has a passage which connects inlet port **232** to the auxiliary port **236**, allowing unrestricted gas flow from the third flow passage to the fifth flow passage **264**. The outlet port **234** is plugged by a blocking seal **430** so that the fourth flow passage **262** is isolated from the third and fifth flow passages **258**, **264**. The pressure transducer component **415** is mounted to the cap component **410** so that it can measure the pressure of the process gas within the third and fifth flow passages **258**, **264**.

(119) Downstream of the cap component **410** and the pressure transducer component **415**, process gas flows through the laminar flow component **420** coupled to the fourth flow component mounting region **240**. Process gas enters the laminar flow component **420** at the inlet port **242**, flows through the restrictor **421**, and out of the laminar flow component **420** through the outlet port **244**. The process gas is then conducted by the fourth flow passage **262** to the gas outlet **260**.

(120) A mass flow controller **800** exhibiting a broad dynamic range of possible mass flow rates is

shown in FIGS. 19 and 20. The mass flow controller **800** incorporates a monolithic base **200**, a control valve component **405**, a pressure transducer component **415**, a cap component **410**, and two laminar flow components **420**. By sizing the restrictors **421** in the laminar flow components **420**, a large dynamic range can be achieved. Selectively enabling the laminar flow component **420** having the appropriate restrictor **421** for the desired mass flow rate allows a single mass flow controller to serve in the place of two individual mass flow controllers having different ranges of mass flow rates, achieving a considerable cost savings.

(121) The mass flow controller **800** positions the cap component **410** and the pressure transducer component **415** in the second flow component mounting location **220**. The auxiliary port **226** of the second flow component mounting location **220** is blocked by a blocking seal **430** which prevents process gas from exiting through the auxiliary port **226**. The first laminar flow component **420** is coupled to the third flow component mounting location **230**. The second laminar flow component **420** is coupled to the fourth flow component mounting location **240**. Both laminar flow components **420** incorporate internal on/off valve components to permit selective engagement of one or both of the laminar flow components **420**. As can be seen, it is possible to reconfigure the component locations to permit the same components to be mounted in different flow component mounting regions as a result of the port and passage configurations. The second, third, and fourth flow component mounting regions **220**, **230**, **240** are multi-function flow component mounting regions that enable flexible arrangement of the components as shown in the mass flow controller **800**. This greatly increases the functionality of the monolithic base **200** and does not require separate customized base designs.

(122) The mass flow controller **900** of FIGS. 21 and 22 is another implementation of the high stability device disclosed in FIGS. 15 and 16. In this embodiment, a control valve component **405**, a volumetric expander component **440**, a pressure transducer component **415**, a cap component **410**, and a laminar flow component **420** are coupled to a monolithic base **200**. The volumetric expander component **440** is coupled to the second flow component mounting location **220** and the pressure transducer component **415** and cap component **410** are coupled to the third flow component mounting location **230**. The auxiliary port **226** of the second flow component mounting location **220** and the outlet port **234** of the third flow component mounting location **230** are blocked with blocking seals **430**. Thus, no process gas flows through the auxiliary passage **256**. The fourth flow passage **262** is also blocked, except for process gas that flows through the laminar flow component **420**.

(123) Turning to FIGS. 23 and 24, a seventh embodiment of the mass flow controller **1000** is disclosed. This embodiment is designed to be a standard mass flow controller with no additional special functionality. The mass flow controller **1000** has a monolithic base **200**, a control valve component **405**, two cap components **410**, a pressure transducer component **415**, and a laminar flow component **420**. The first cap component **410** is coupled to the second flow component mounting region **220** and the upper port on the first cap component **410** is plugged because there is no pressure transducer component **415** installed on the first cap component **410**. The auxiliary port **226** of the second flow component mounting region **220** is blocked by a blocking seal **430** to prevent flow into the auxiliary passage **256**.

(124) The second cap component **410** is coupled to the third flow component mounting region **230** and the pressure transducer component **415** is attached to the second cap component **410**. This enables the pressure transducer component **415** to sample the pressure upstream of the laminar flow component **420**. In this embodiment, the outlet port **234** of the third flow component mounting region **230** is blocked by a blocking seal **430** to prevent flow upstream of the laminar flow component **420** from entering the fourth flow passage **262** without first passing through the laminar flow component **420**. It is possible that the pressure transducer component **415** could be mounted to the first cap block **410** instead of the second cap block **410** if so desired, as the pressure in the internal passages of each of the first and second cap blocks **410** is equal.

(125) Other mass flow controllers may be assembled with yet further variations in flow components. For example, it is within the scope of the invention to incorporate the features of one or more of the individual components into a single combination component. Specifically, a hybrid control valve component and pressure transducer component may be created which simultaneously controls process gas flow into the mass flow controller while simultaneously measuring the pressure of the gas downstream of the valve. This may free up additional flow component mounting locations for a more compact installation or the inclusion of additional flow components.

(126) The monolithic bases **100**, **200** may be used in a variety of installations in a substantially identical configuration. Monolithic bases are considered to be substantially identical even though particular blocks may vary due to minor imperfections, normal manufacturing tolerances, variations in flow component mounting arrangements, etc. Monolithic bases are also considered substantially identical where they have the same port arrangement and flow passage arrangement, even though the exact dimensions of the flow ports may vary. Where the bases have different port and passage variations, they are not deemed to be substantially identical.

(127) Mass flow controller operating characteristics typically include maximum mass flow rate, minimum mass flow rate, mass flow rate supply accuracy, dynamic operating range, startup response time, and shut-off response time. As noted above, a variety of configurations of flow components can achieve different operating characteristics. Though altering the restrictor in a laminar flow component can achieve different ranges of maximum and minimum flow rates, the dynamic operating range is limited when a single laminar flow component is used. Thus, it is often advantageous to add a second laminar flow component having a different restrictor to increase the dynamic operating range. In yet other embodiments, more than two laminar flow components may be added to further enhance the dynamic operating range.

(128) In other embodiments, it may be desirable to change the flow components to incorporate more pressure transducers, different components such as bleed components or volumetric expanders, or yet other flow components. Thus, a wide range of types of flow components may be incorporated into a single mass flow controller. Alternately, the flow components may be differently mounted on the same monolithic base to provide a variety of configurations.

(129) Furthermore, temperature sensor components may be incorporated into the monolithic base, the control valve component, or any of the other components within the system. This enables an electronic control element to compensate for the temperature of the process gas and further enhance system accuracy. The electronic control element operates the valves and measures temperature and pressure to obtain the desired mass flow rates. The electronic control element is also capable of networked communication with other electronic devices in the system, so that it can send and receive data such as pressures in a process manifold downstream from the mass flow controller or instructions to start, stop, or alter the commanded mass flow rate. The electronic control element also stores all system calibration data to ensure that parameters such as the characterization data of the restrictor(s) in the laminar flow components.

(130) Dynamic operating ranges for mass flow controllers having a single laminar flow component may be in the range of 20:1. Dynamic operating ranges for mass flow controllers having two laminar flow components may be as high as 400:1. Dynamic operating ranges for mass flow controllers having three laminar flow components may be as high as 8000:1. Each additional laminar flow component can increase the dynamic operating range by 20 times when the restrictors are selected appropriately.

(131) Furthermore, achieving the desired mass flow rate supply accuracy can require additional flow components. Though a mass flow controller incorporating a single laminar flow component may have a high accuracy within a portion of the dynamic operating range, its accuracy may not be constant throughout the dynamic operating range. Accuracy may be enhanced by incorporating multiple laminar flow components or other flow components. Target mass flow rate supply accuracy may be 1%, 0.9%, or 0.5%.

(132) The startup and shut-off response time of the unit is also affected by the volume between the control valve component and the one or more laminar flow components. Thus, if extremely fast response times are required, it may be necessary to minimize this volume. Alternately, if high stability is desired, a larger volume may be desirable to dampen pulses in the supplied mass flow rate. The volume may be altered to achieve a desired stability. The stability of the mass flow rate may be measured as a peak percent error from the desired mass flow rate. Target stability may be 1%, 0.8%, 0.5%, or even 0.25%.

(133) Finally, the incorporation of bleed components may provide higher accuracy at extremely low flow rates. The bleed component may be constructed similarly to a laminar flow component with a different orifice or restrictor provided, and it may or may not incorporate a valve to control the opening of the orifice or restrictor.

(134) The present invention may also be a process for manufacturing semiconductor devices incorporating a mass flow controller. This process may incorporate any system where a mass flow controller is connected to a gas supply and a controlled mass flow rate of a process gas is delivered to the process. Some representative semiconductor processes may include deposition, removal, patterning, or modification of electrical properties. Deposition processes may include physical vapor deposition, chemical vapor deposition, electrochemical deposition, molecular beam epitaxy, and atomic layer deposition. Removal may include wet or dry etching and chemical-mechanical planarization. Patterning includes lithography processes which may incorporate deposition of photoresist and plasma ashing. Modification of electrical properties may include processes for doping by diffusion or ion implantation, or annealing by furnace annealing or rapid thermal annealing. The invention may include equipment for processing semiconductor devices through any process requiring controlled gas flow.

(135) The present invention may also be a process for allowing a customer to specify and construct a mass flow system integrating off the shelf components into the customer's semiconductor manufacturing equipment. Furthermore, additional components may be purchased so as to enable reconfiguration of existing equipment at lower cost, rather than purchasing a new mass flow controller or mass flow control system for a new application. Individual components may be substituted or reconfigured to achieve different control objectives.

Section II

(136) The present invention is directed to a method of improving the transient turn on performance of pressure based apparatuses for controlling mass flow. These apparatuses are used to provide steady state control of gas flows in a variety of industrial applications. In some embodiments, these apparatuses may be mass flow controllers which control the mass flow rate of a gas. Semiconductor fabrication is one industry which demands high performance in control of gas flows. As semiconductor fabrication techniques have advanced, customers have recognized the need for rapid and repeatable transient response in starting a gas flow. The need to reduce process times, minimize wasted process gas, improve yield, and increase factory throughput all drive the need for improved transient responses in apparatuses for controlling flow. In particular, the transient turn on time, or "TTO time," is a key parameter of next generation apparatuses for controlling flow. Lower transient turn on times and settling times help to drive down semiconductor fabrication costs.

(137) FIG. 25 shows a cross sectional view of an apparatus for controlling flow **100A**. The gas flow path of the process gas is indicated by arrows which illustrate the path that the process gas takes through the apparatus **100A**. The apparatus **100A** has a base **102A** comprising a gas inlet **104A** and a gas outlet **110A**, the gas flow path extending between the gas inlet **104A** and the gas outlet **110A**. A supply of process gas is delivered to the gas inlet **104A** of the base **102A**. The process gas then flows through a proportional valve **120A** into a P1 volume **106A** within the base **102A**. The proportional valve **120A** meters the amount of process gas which passes to the P1 volume **106A**. The proportional valve **120A** is capable of providing proportional control of the process gas such that it need not be fully open or closed, but instead may have intermediate states

to permit control of the mass flow rate of process gas. In some embodiments, the proportional valve **120A** may have a closure member which can move to a plurality of distinct positions between the fully open and fully closed positions. In some cases, the closure member may be infinitely adjustable. In yet other embodiments, the proportional valve **120A** may open and close rapidly to control the amount of gas which flows through the proportional valve **120A**. The proportional valve **120A** may utilize any method of metering the mass flow rate of process gas into the P1 volume **106A**.

(138) A pressure transducer **130A** is attached to the base **102A** and is fluidly coupled to the P1 volume **106A** so that it can sample the pressure within the P1 volume **106A**. The base **102A** may incorporate one or more additional ports **112A** to permit alternate configurations. In the present embodiment, the port **112A** is blocked with a cap component **140A**. Alternate configurations may incorporate additional components or position the components differently to achieve different mass flow rates, or additional functions to further improve transient performance.

(139) Next, the process gas flows out of the P1 volume **106A** into an on/off valve **150A**. Internal to the on/off valve **150A** is a valve seat **152A** and a closure member **154A**. When the apparatus **100A** is delivering process gas, the on/off valve **150A** is in an open state, such that the valve seat **152A** and the closure member **154A** are not in contact. This permits free flow of the process gas, and provides a negligible restriction to fluid flow. When the apparatus **100A** is commanded to stop the flow of the process gas, the closure member **154A** and the valve seat **152A** are biased into contact by the spring **156A**, stopping the flow of process gas through the on/off valve **150A**.

(140) Downstream of the valve seat **152A** is a characterized flow restrictor **160A** which provides a known restriction to fluid flow. This restriction may be described as a flow impedance, a higher flow impedance providing an increased restriction to fluid flow. The characterized flow restrictor **160A** may be selected or adjusted to have a range of flow impedances. This allows the same apparatus **100A** to be optimized for different ranges of mass flow rates that the apparatus **100A** may supply. The characterized flow restrictor **160A** may be formed as a porous block, a device having small orifices or channels, or any other means of providing a restriction to process gas flow that is characterized across a target dynamic operating range of mass flow rates. The characterized flow restrictor **160A** has a greater flow impedance than the passages upstream and downstream of the characterized flow restrictor **160A**. After passing through the characterized flow restrictor **160A**, the process gas exits the gas outlet **110A**. Generally, the gas outlet **110A** of the apparatus **100A** is coupled to a manifold, the manifold directing a plurality of process gases to an applicator in the process equipment.

(141) Optionally, temperature sensors may be employed to further enhance the accuracy of the apparatus **100A**. A temperature sensor **114A** is shown in FIG. 25, located within the base **102A** so that it can measure the temperature near the P1 volume **106A**. Additional temperature sensors **114A** may be employed in a variety of locations, including the proportional valve **120A**, the pressure transducer **130A**, and the on/off valve **150A**. Furthermore, a pressure sensor downstream of the characterized flow restrictor **160A** may be utilized to further increase the accuracy of the mass flow delivered by the apparatus **100A**.

(142) Turning to FIG. 26, a block diagram illustrates the control system for the apparatus **100A** of FIG. 25. This block diagram shows a controller **400A** comprising an apparatus controller **200A** and a system controller **300A**. In this embodiment, the controller **400A** is divided into two parts. The apparatus controller **200A** provides all control functions for the apparatus **100A**. The apparatus controller **200A** has a communication interface **210A**, a proportional valve controller **220A**, a pressure transducer interface **230A**, an on/off valve controller **250A**, a temperature sensor interface **260A**, a processor **270A**, and memory **280A**. The communication interface **210A** is configured to provide a communications link between the apparatus controller **200A** and the system controller **300A**. Optionally, the temperature sensor interface **260A** may be omitted if the additional accuracy provided by the temperature sensor **114A** is not required.

(143) The system controller **300A** has a corresponding communication interface **310A**, a processor **370A**, and memory **380A**. The system controller **300A** coordinates all high level functions necessary to perform the desired process. The communication interface **310A** of the system controller **300A** sends and receives commands through a communications bus **390A**. The communications bus **390A** connects to the communication interface **210A** of the control module **210A** of the apparatus **100A**. The communications bus **390A** may connect to a single apparatus controller **200A**, or it may connect to a plurality of apparatus controllers **200A**, each apparatus controller **200A** operating a distinct apparatus **100A**. Not all apparatus controllers **200A** need control an apparatus **100A** for controlling mass flow. Instead, other types of process equipment may also be controlled. Furthermore, there may be a plurality of communications buses **390A** to connect all the devices required to perform the desired process.

(144) Internal to the system controller **300A**, the processor **370A** and the memory **380A** operate to carry out the desired process. The processor **370A** provides the timing necessary to ensure that the appropriate steps are carried out for the desired duration, and provides instructions to the apparatus controller **200A** of the apparatus **100A**. The necessary information is transmitted from the system controller **300A** to the apparatus controller **200A** as a gas flow activation signal. The gas flow activation signal may consist of information such as the desired state of the apparatus **100A** (i.e. flowing gas or not flowing gas), a predetermined mass flow rate needed to complete the process, a predetermined mass flow rate required at a future time, and the future time at which the predetermined mass flow rate is required. In other embodiments, the gas flow activation signal may provide information that instructs the apparatus controller **200A** to begin flowing gas upon receipt of a trigger signal.

(145) A time stamp or other synchronization method may be provided in the instructions to ensure that the system controller **300A** and the apparatus controller **200A** are synchronized. This ensures that process events occur at the desired time. Other methods may be used to ensure that the apparatus begins flowing gas at the desired time. In addition, other signals may be transmitted between the system controller **300A** and the apparatus controller **200A**. For instance, an acknowledgement message or current status message may be issued from the apparatus controller **200A** to the system controller **300A** to provide the current state or confirm the receipt of instructions. Status messages may be provided automatically, in response to an input, or in response to a polling message from the system controller **300A**. Where a trigger signal is used instead of a future turn-on time, the apparatus controller **200A** will begin priming the apparatus **100A** and will wait for the trigger signal to open the on/off valve **150A**.

(146) Similarly, the processor **270A** and memory **280A** of the apparatus controller **200A** operate to maintain timing, send and receive messages through the communication interface **210A**, and operate the functions of the apparatus **100A**. The processor **270A** of the apparatus **100A** implements a closed loop feedback system. When the apparatus controller **200A** is instructed to deliver process gas, the apparatus controller **200A** monitors the pressure in the P1 volume **106A** using the measurements taken from the pressure transducer **130A**. The pressure transducer interface **230A** takes readings from the pressure transducer **130A**. This information, in combination with the known flow impedance provided by the characterized flow restrictor **160A**, is used to calculate the mass flow rate of the process gas through the apparatus **100A**. A temperature value determined from the temperature sensor **114A** may also be used to further enhance the accuracy of the calculation. The value of the flow impedance is stored in the memory **280A** along with other constants and calibration data to enable accurate calculation of the various process parameters.

(147) The processor **270A** then compares the current mass flow rate through the apparatus **100A** against the predetermined mass flow rate provided by the system controller **300A**. The proportional valve controller **220A** commands the proportional valve **120A** to increase or decrease the flow rate of process gas into the P1 volume **106A** to achieve a target pressure in the P1 volume **106A** that will result in the predetermined mass flow rate. This process is continually repeated until the

apparatus controller **200A** is commanded to stop delivering process gas. At this time, the proportional valve **120A** is closed. The on/off valve controller **250A** also commands the on/off valve **150A** to close, halting flow of the process gas through the outlet **110A**. The on/off valve **150A** remains closed until the apparatus controller **200A** is instructed to deliver process gas. At that time, the on/off valve **150A** is opened and the apparatus **100A** resumes operation.

(148) In yet other embodiments, the controller **400A** may incorporate the functionality of both the apparatus controller **200A** and the system controller **300A** into a single device which need not be connected by a communications bus **390A**. Instead, the proportional valve **120A**, on/off valve **150A**, and other elements are interfaced directly by a single controller which generates a gas flow activation signal internally to the controller **400A**. A single controller may interface more than one device. This configuration has the advantage of elimination of redundant hardware, but requires greater controller complexity.

(149) In operation, the apparatus controller **200A** is instructed to begin delivering flow at a future time when the gas flow activation signal is provided. This generally occurs when the apparatus **100A** is shut off and no gas is being delivered to the gas outlet **110A**. The gas flow activation signal is generated by the system controller **300A** and received by the apparatus controller **200A**. The gas flow activation signal includes information which instructs the apparatus controller **200A** to change from zero flow to a predetermined flow rate having a non-zero positive value. The receipt of the gas flow activation signal by the apparatus controller **200A** begins a priming period. Upon receipt of the gas flow activation signal, the apparatus **100A** prepares to deliver gas at the predetermined flow rate. In this instance, “upon” means at any time concurrent or subsequent to the event. Thus, the apparatus **100A** may prepare to deliver gas at any time concurrent or subsequent to the receipt of the gas flow activation signal by the apparatus controller **200A**.

(150) When the apparatus **100A** is shut off, both the proportional valve **120A** and the on/off valve **150A** are closed. In order to prime the apparatus prior to beginning the flow of gas, the proportional valve controller **220A** of the apparatus controller **200A** commands the proportional valve **120A** to open to achieve a target pre-flow pressure in the P1 volume **106A**. The target pre-flow pressure is calculated to achieve the predetermined flow rate based on the flow impedance of the flow restrictor **160A**, subsequent to the opening of the on/off valve **150A**. At the occurrence of the turn on time $t_{sub.2}$ at which the gas flow activation signal commands the gas flow to begin, the on/off valve **150A** is moved from an off state where the on/off valve **150A** is closed to an on state where the on/off valve **150A** is open. This ends the priming period for the apparatus **100A**. The gas then begins to flow out of the P1 volume **106A**, past the on/off valve **150A** and the flow restrictor **160A**, and out of the gas outlet **110A**. The apparatus **100A** then drives the delivered flow rate to the predetermined flow rate using its normal control system. This system typically operates on a PID feedback loop to ensure that a delivered flow rate is substantially identical to the predetermined flow rate.

(151) The transient turn on time of the apparatus **100A** is measured from the time that the on/off valve **150A** is commanded to open until the delivered flow rate delivered by the apparatus **100A** has stabilized within a certain range. In many instances, the delivered flow rate must be within plus or minus 2% of the predetermined mass flow rate. However, stability windows of plus or minus 5%, 1%, 0.8%, 0.5%, 0.25%, or 0.1% may also be specified, depending on the process requirements.

(152) Turning to FIG. 27, a schematic diagram of the apparatus **100A** discussed above is provided. When the apparatus **100A** is instructed to deliver process gas, the process gas flows through the proportional valve **120A**. The proportional valve **120A** meters the process gas into the P1 volume **106A**. Process gas then passes through the on/off valve **150A** and the characterized flow restrictor **160A**. Thus, a known mass flow of process gas is delivered to the process. When the apparatus **100A** is instructed to cease delivery of process gas, the proportional valve **120A** and the on/off valve **150A** are closed, stopping flow of process gas.

(153) Typical commercially available apparatuses for flow control provide a transient turn on time in the range of 500 to 1000 milliseconds with an accuracy of plus or minus 2%. The semiconductor industry typically uses a range of plus or minus 2% of the set point as the window for measuring the transient turn on time. The transient turn on time is determined by the earliest time that the delivered mass flow rate enters and remains within the 2% window. Although other percentages may be used, many semiconductor manufacturers adhere to the 2% specification.

(154) The transient turn on time is dictated by inherent limitations in flow sensing and the speed that the proportional valve **120A** can respond to commands. Flow sensing limitations are controlled by the frequency at which the pressure transducer **130A** reading is taken and the speed at which the pressure transducer **130A** can respond to changes in pressure in the P1 volume. The proportional valve **120A** also has limitations on how fast it can modulate its opening position or otherwise control the metered flow rate into the P1 volume **106A**. Furthermore, the size of the P1 volume **106A** also affect the transient turn on time, the transient turn off time, and the stability of the resulting mass flow. Faster transient turn on and turn off times may be achieved by minimizing the size of the P1 volume **106A**, but there are limitations to this approach. For instance, stability may be adversely affected by minimizing the size of the P1 volume **106A**.

(155) Thus, commercially available apparatuses are unable to significantly reduce their transient turn on times below 500 milliseconds due to the inherent system limitations of pressure based apparatuses. These apparatuses accept instructions from the system controller **300A** that essentially consist of a command to change the predetermined mass flow rate from zero to a given value, with no advance notice of the predetermined mass flow rate. Accordingly, the transient turn on time is merely the time to achieve the predetermined mass flow rate in a single instantaneous step. Current commercially available apparatuses receive no advance notice of the turn on command.

(156) The present approach does not rely on the use of extremely fast proportional valves, on/off valves, or pressure transducers to achieve substantial reductions in transient turn on times. For instance, commercially available on/off valves may have response times in the range of 3 to 80 milliseconds. On/off valves having a response time of 50 milliseconds are commonly available at reasonable prices. Faster valves can be used, but generally incur additional cost. For the sake of discussion, a response time of 50 milliseconds is assumed.

(157) In order to achieve improved transient turn on times, the gas flow activation signal issued by the system controller **300A** includes information about both the predetermined mass flow rate required by the process and a future turn on time $t_{sub.2}$ that that apparatus **100A** should open the on/off valve **150A** to begin supplying process gas. The apparatus controller **200A** receives the gas flow activation signal at a first time $t_{sub.1}$. Providing the gas flow activation signal in advance of the turn on time $t_{sub.2}$ allows the apparatus **100A** to pre-pressurize the P1 volume **106A** during the priming period and overcomes limitations related to valve and pressure transducer response times. The turn on time $t_{sub.2}$ is also known as a second time. In alternate embodiments, the gas flow activation signal includes information instructing the apparatus controller **200A** to wait for a trigger signal to open the on/off valve **150A** rather than a specific time for beginning gas flow.

(158) In the first method, shown in FIG. **28**, the apparatus controller **200A** of the apparatus **100A** waits for the gas flow activation signal instructing it to deliver a predetermined mass flow rate at future turn on time $t_{sub.2}$. During the time prior to the receipt of the command, the apparatus **100A** is generally off. However, it is possible that the same method may be used to alter the predetermined flow rate from one non-zero mass flow rate to another non-zero mass flow rate. Upon receipt of the gas flow activation signal, the apparatus controller **200A** computes a target pre-flow pressure in the P1 volume **106A** to achieve the predetermined mass flow rate. The proportional valve **120A** is then opened under proportional integral derivative (“PID”) control to achieve the target pre-flow pressure in the P1 volume **106A**. Once the target pre-flow pressure in the P1 volume **106A** is reached, the proportional valve **120A** is closed to prevent the pressure in the P1 volume **106A** from overshooting the target pressure. At the turn on time $t_{sub.2}$, the on/off valve

150A is opened and the apparatus **100A** begins delivering process gas. The proportional valve **120A** is then opened to maintain the pressure in the P1 volume **106A**.

(159) Depending on the duration of the priming period and the speed at which the proportional valve **120A** is able to reach the target pre-flow pressure in the P1 volume **106A**, the proportional valve **120A** may not close before the turn on time $t_{sub.2}$ is reached. In this case, it is likely that the proportional valve **120A** will meter less process gas into the P1 volume **106A** than needed to achieve the target pre-flow pressure. The apparatus **100A** is now in a primed condition which is selected to achieve the predetermined flow rate. In the ideal implementation, adequate priming period is provided so that the proportional valve **120A** is able to pre-pressurize the P1 volume **106A** prior to the turn on time $t_{sub.2}$. In some other embodiments, the turn on time $t_{sub.2}$ may occur prior to achievement of the target pre-flow pressure or immediately upon achievement of the pre-flow pressure. It is not necessary that the proportional valve **120A** close prior to the turn on time $t_{sub.2}$. It is conceived that it remains open, achieving the target pre-flow pressure in the P1 volume **106A** exactly at the turn on time $t_{sub.2}$.

(160) In order to achieve the target pre-flow pressure in the P1 volume **106A**, the proportional valve **120A** may open fully, or may only open partially. The opening position and the pressurization profile of the pressure in the P1 volume **106A** may be adjusted in any manner necessary to achieve the target pressure. The pressurization profile may be controlled so that pressure rises linearly. Or in other embodiments, the pressurization profile may be controlled so that the P1 volume **106A** reaches the target pre-flow pressure as soon as possible without overshoot.

(161) In some embodiments, the estimated time to achieve the primed condition may be calculated, and the proportional valve **120A** may be opened earlier or later to alter the time required to prime the P1 volume **106A**. In some instances, the slope may vary depending on the predetermined flow rate, the pressure of the supplied process gas at the gas inlet **104A**, the priming period, or other factors.

(162) The present method offers the advantage of eliminating the need to pressurize the P1 volume **106A** subsequent to the turn on time $t_{sub.2}$. This reduces the transient turn on time, particularly for low flow rates. The apparatus **100A** may be arranged such that the on/off valve **150A** is upstream or downstream of the calibrated flow restrictor **160A**. In the event that the calibrated flow restrictor **160A** is upstream of the on/off valve **150A**, a pulse of high pressure process gas is delivered before the flow stabilizes. This undesirably wastes process gas, but the pulse will not meaningfully impact the transient turn on time, as it occurs much more rapidly than the flow rate can stabilize within the target boundaries. In the event that the calibrated flow restrictor **160A** is downstream of the on/off valve **150A**, no pulse occurs, but the process gas must flow through the flow restrictor **160A**. Furthermore, there is a small additional unpressurized volume downstream of the on/off valve **150A** that results in a slight pressure drop. In practice, this additional volume causes a negligible change in pressure in the P1 volume **106A** as long as the flow restrictor **160A** is located near the valve seat **152A**. In some embodiments, the flow restrictor **160A** is located adjacent the valve seat **152A**, with 1 cc or less volume between the flow restrictor **160A** and the valve seat **152A**. A volume of 0.5, 0.2, 0.1, or 0.02 cc is preferred.

(163) The present method operates most effectively where the predetermined mass flow rate is low as compared with the mass of process gas in the P1 volume **106A**. In contrast to other methods of improving response time, this method offers the dual advantages of improving stability and reducing transient turn on times. This is because the P1 volume **106A** need not be reduced in size to the utmost degree, lessening the burden on the proportional valve **120A** and the apparatus controller **200A**. However, for larger mass flow rates as compared with the mass in the P1 volume **106A**, the mass flow rate dips undesirably as the process gas flows out of the P1 volume **106A**. The P1 volume **106A** is generally of insufficient size to act as a cushion when the on/off valve **150A** is opened. Then, the proportional valve **120A** is commanded to open rapidly. Both the apparatus controller **200A** and the proportional valve **120A** have speed limitations which undesirably

lengthen the time before the target pressure in the P1 volume is restored. Though the present method still offers advantages over commercially available apparatuses, this method does not provide similar stability and transient turn on performance over the entire operating range.

(164) Having a larger P1 volume **106A** to mass flow rate results in an improved damping effect, preventing the pressure in the P1 volume **106A** from dropping rapidly after the turn on time $t_{sub.2}$. Increasing mass flow rates increase the burden on the apparatus **100A**. As noted previously, the proportional valve **120A** generally closes prior to the turn on time $t_{sub.2}$ as a result of the need to avoid overshooting the target pressure in the P1 volume **106A**. Once the turn on time $t_{sub.2}$ passes, the proportional valve **120A** is instantaneously commanded to open to counteract the rapid change in pressure in the P1 volume **106A** resulting from the opening of the on/off valve **150A**. Thus, the ability of the apparatus **100A** to maintain the predetermined mass flow rate becomes increasingly dependent on the response time of the proportional valve **120A** and the control loop implemented in the apparatus controller **200A** as the predetermined mass flow rate increases.

(165) FIG. **29** shows the test results of the method of FIG. **28** for a range of predetermined mass flow rates. Mass flow rates of 150, 300, 750, 1000, 1500, and 1800 standard cubic centimeters per minute (“sccm”) were tested at a process gas supply pressure of 515 kpa (60 psig). The mass flow rates were graphed from the turn on time $t_{sub.2}$ to 500 milliseconds and normalized so that the predetermined mass flow rate was equal to 100%. As the predetermined mass flow rates increased, the delivered flow rate drops to a lower percentage of the predetermined mass flow rate, with the 1000 sccm, 1500 sccm, and 1800 sccm mass flow rates falling below 92% of the desired rate. Furthermore, the time to reach 2% of the predetermined mass flow rate was approximately 500 milliseconds for the highest flow rates.

(166) In yet further embodiments, a bleed valve may be used, such as that shown in the apparatus **500A** of FIG. **30A**. The bleed valve **170A** is capable of bleeding gas from the P1 volume **106A** into a non-process location through a bleed passage **182A** to a bleed outlet **180A**, the bleed outlet **180A** being isolated from the gas outlet **110A**. The bleed valve **170A** may be a proportional valve similar to the proportional valve **120A**. This allows the pressure in the P1 volume **106A** to be controlled by the bleed valve **170A** while simultaneously controlling the flow of gas through the proportional valve **120A** into the P1 volume **106A**. This modification provides greater control over the resulting pre-flow pressure and the flow rate through the proportional valve **120A**. The bleed valve **170A** may be controlled in any manner desired to achieve the desired pre-flow pressure, including simultaneous opening with the proportional valve **120A**, subsequent opening, or even opening prior to the opening of the proportional valve **120A**. The bleed valve **170A** may be opened with a variety of profiles and ramp rates to control the pressure in the P1 volume **106A** including linear, exponential, or other profiles. It is also contemplated that the opening of the proportional valve **120A** may be held constant while the bleed valve **170A** controls the pressure in the P1 volume **106A**. In other embodiments, such as the one shown in FIG. **30B**, the bleed valve **170A** may be an on/off valve similar to the on/off valve **150A**. The bleed valve **170A** is used in combination with a second characterized restrictor **190A** located within the bleed passage **182A**. This allows a known flow rate of gas to be metered out of the P1 volume **106A** to the bleed outlet **180A** and the non-process location. Alternately, the second characterized restrictor **190A** may be located between the bleed valve **170A** and the P1 volume **106A**.

(167) FIG. **31** illustrates a second method of improving transient turn on time for the apparatus **100A**. Initially the apparatus **100A** is off, such that it delivers no process gas. The apparatus controller **200A** of the apparatus **100A** waits for the gas flow activation signal instructing it to deliver a predetermined mass flow rate of gas at the future turn on time $t_{sub.2}$. Once the signal is received, the priming period begins. The apparatus controller **200A** calculates the target pre-flow pressure in the P1 volume **106A** required to achieve the predetermined mass flow rate. The proportional valve **120A** is then opened under PID control, with the goal of achieving the target pre-flow pressure in the P1 volume **106A** at the turn on time $t_{sub.2}$. Thus, two boundary conditions

are specified. Not only is the pressure driven to the target pre-flow pressure by the proportional valve **120A**, but the target pre-flow pressure is reached at the time that the on/off valve **150A** is scheduled to open. Then the on/off valve **150A** is opened at the turn on time $t_{sub.2}$. Finally, the proportional valve **120A** is controlled so that the pressure in the P1 volume **106A** is maintained at the target pressure. This causes the delivered flow rate of the apparatus **100A** to quickly settle at the predetermined mass flow rate. As mentioned above, an apparatus incorporating the bleed valve (168) FIG. **32** discloses a third method of improving transient turn on time for the apparatus **100A**. In this method, the proportional valve **120A** is driven such that the target pre-flow pressure is achieved in the P1 volume **106A** at the turn on time $t_{sub.2}$ as with the second method disclosed above. Simultaneously, the mass flow rate across the proportional valve **120A** is also driven to the predetermined mass flow rate at the turn on time $t_{sub.2}$. This offers a further improvement because three boundary conditions are met simultaneously. By controlling the mass flow rate across the proportional valve **120A** at the turn on time $t_{sub.2}$, the delivered mass flow rate through the apparatus **100A** will settle to very near the predetermined mass flow rate. Furthermore, the proportional valve **120A** will not need to move significantly to maintain the pressure in the P1 volume at the target pressure.

(169) Thus, this method has the goal of priming the P1 volume **106A** to the target pre-flow pressure, but it also has the goal of having the proportional valve **120A** open and flowing approximately the predetermined mass flow rate into the P1 volume **106A** at the turn on time $t_{sub.2}$. This has the benefit of minimizing the necessary movement of the proportional valve **120A** upon the opening of the on/off valve **150A**. Therefore, the corrective action commanded by the proportional valve **120A** PID loop is reduced such that the delivered flow through the characterized flow restrictor **160A** does not drop out of the target range. Furthermore, the performance of the apparatus **100A** is nearly identical for a wide range of predetermined mass flow rates. The present method is able to settle far faster than the method of FIG. **28** at all mass flow rates.

(170) The mass flow into the P1 volume **106A** can be calculated during the priming period. During this period, the on/off valve **150A** is closed. The volume of the P1 volume **106A** is measured and stored in the memory **280A** of the apparatus controller **200A**. The mass flow into the P1 volume **106A** is calculated using the measured pressure from the pressure transducer **130A** and the current temperature combined with the time based derivative of the Ideal Gas law. Thus, the calculation of mass flow into the P1 volume **106A** is:

$$\text{Massflow(sccm)} = dP(\text{atm})/dt(\text{min}) * V(\text{cc}) * \text{Temp.sub.reference}(\text{deg K}) / \text{Temp.sub.current}(\text{deg K})$$

(171) Thus, the mass flow in sccm is equal to the change in pressure in atm per time in minutes multiplied by the volume of the P1 volume **106A** times the reference temperature in degrees Kelvin divided by the current temperature in degrees Kelvin. In the semiconductor industry, the reference temperature is defined to be 273.15° Kelvin (0° C.). The change in pressure is the pressure drop from the gas source to the P1 volume **106A**, as regulated by the proportional valve **120A**. The apparatus controller **200A** can then adjust the rate of increase of pressure in the P1 volume **106A** to achieve the three boundary conditions of target pressure and mass flow rate into the P1 volume **106A** at the turn on time $t_{sub.2}$. By controlling the mass flow rate into the P1 volume **106A**, the position of the proportional valve **120A** is controlled. This is because the mass flow rate across the proportional valve **120A** is determined by the position of the closure member of the proportional valve **120A**. In yet other embodiments, the apparatus controller **200A** only drives the mass flow rate into the P1 volume **106A** at the turn on time $t_{sub.2}$ without achieving the pre-flow pressure at the turn on time $t_{sub.2}$. This can allow even faster transient turn on time performance at the expense of some overshoot during initial opening of the on/off valve **150A**.

(172) The exact method of reaching the boundary conditions may vary. For instance, the pressurization profile may have a linear ramp of pressure in the P1 volume **106A** during the priming period. Alternately, a non-linear profile, a plurality of linear ramps, or other profiles may be used. In yet other embodiments, the target pressure may be sculpted to achieve the

predetermined mass flow rate into the P1 volume **106A** through the use of a variety of profiles. This may include combinations of several different curves to achieve the desired pressurization profile. Depending on the commanded mass flow rate and the amount of time available, a wide range of profiles may be employed. In some embodiments, the pressurization profile or the time to complete the pressurization profile may be constant regardless of the predetermined flow rate. In yet other embodiments, the pressurization profile or the time to complete the pressurization profile may vary based on the predetermined flow rate or the time available before the turn on time $t_{sub.2}$.

(173) In one very simple pressurization profile, it is conceived that the proportional valve **120A** is opened to flow a constant mass flow rate equal to the predetermined mass flow rate through the apparatus **100A**. Thus, the P1 volume **106A** is simply filled at a constant mass flow rate and the pressure in the P1 volume **106A** is represented by the area under the mass flow rate profile. The proportional valve **120A** must then be opened at a calculated time in advance of the turn on time $t_{sub.2}$ in order to reach the desired pressure in the P1 volume. For greater predetermined mass flow rates, this advance time must decrease. Thus, this algorithm requires significant advance notice of the turn on time $t_{sub.2}$ in order to operate effectively for all predetermined mass flow rates. Furthermore, decreasing the mass flow rates used to fill the P1 volume **106A** during the priming period further increases the amount of time required prior to the turn on time $t_{sub.2}$.

(174) In alternate profiles, it is conceived that the proportional valve **120A** may be opened such that the mass flow rate into the P1 volume **106A** progressively increases in advance of the turn on time $t_{sub.2}$. For small predetermined mass flow rates, the proportional valve **120A** may be commanded to open such that the mass flow rate across the proportional valve **120A** increases as the turn on time $t_{sub.2}$ approaches. The slope of the pressurization profile may increase for greater predetermined mass flow rates because the P1 volume **106A** pressurizes more quickly at greater predetermined mass flow rates. It is not necessary that the mass flow rate across the proportional valve **120A** be limited to constant or increasing slope. Instead, it is conceived that for some predetermined mass flow rates, the proportional valve **120A** is commanded to deliver a mass flow rate greater than that required to achieve the predetermined mass flow rate. This will pressurize the P1 volume **106A** even more quickly than a constant slope pressurization profile. Then, the mass flow rate across the proportional valve **120A** can be gradually reduced. The mass flow rate across the proportional valve **120A** can end at the predetermined mass flow rate when the turn on time $t_{sub.2}$ elapses.

(175) As can be seen, the flow rate across the proportional valve **120A** may be linear, exponential, or any other profile required to achieve the desired boundary conditions. Furthermore, the pressure in the P1 volume **106A** is always equal to the area under the mass flow rate profile used to pressurize the P1 volume **106A**. Thus, although the target pre-flow pressure in the P1 volume **106A** may be achieved by opening the proportional valve **120A** to enable a slow bleed, this would not position the proportional valve **120A** such that it delivers the predetermined mass flow rate after the opening of the on/off valve **150A**. Accordingly, a significant correction of the proportional valve **120A** position would be required, lengthening transient turn on time and adversely affecting stability of the delivered mass flow rate through the apparatus.

(176) A plurality of profiles may be employed to optimize the transient turn on time. In yet other embodiments, a plurality of different pressurization profiles may be employed for a range of different predetermined mass flow rates. This can minimize the required priming period before the transient turn on time $t_{sub.2}$. Thus, the behavior of the apparatus **100A** may be optimized for a wide range of process requirements. The accuracy of the delivered flow, the transient turn on time, and the minimum priming period may all be tuned using customized pressurization profiles.

(177) Test results for the apparatus **100A** implementing one embodiment of the method of FIG. 32 are shown in FIG. 33. This method employs a linear ramp of the mass flow rate across the proportional valve **120A** to reach the predetermined mass flow rate through the proportional valve **120A** at the turn on time $t_{sub.2}$. This method delivers greatly improved transient turn on

performance as compared with commercially available apparatuses. Predetermined mass flow rates of 150 sccm, 1000 sccm, and 1800 sccm were tested on an apparatus **100** rated at 1500 sccm. Process gas supply pressures of 480 kpa (55 psig) and 549 kpa (65 psig) were tested for all predetermined mass flow rates. As can be seen, the delivered mass flow rate rises rapidly to the predetermined mass flow rate for all tested flow rates and supply pressures. It overshoots the predetermined flow rate, then falls slightly before settling out. The worst droop occurs with the 1800 sccm, 55 psi test. However, this drops only approximately 1% of the predetermined flow rate, and all of the transient turn on times are significantly less than 100 milliseconds for a stability window of plus or minus 2%. Settling times are also greatly improved, regardless of process gas supply pressure. The method of FIG. **32** delivers transient turn on time which far exceeds that of commercially available apparatuses.

(178) FIG. **34** discloses a fourth method of improving transient turn on time. This method utilizes one of the apparatuses **500A**, **600A** shown in FIGS. **30A** and **30B**. In these methods, the proportional valve **120A** is driven such that the target pre-flow pressure is achieved in the P1 volume **106A** in response to the gas flow activation signal. Simultaneously, the mass flow rate across the proportional valve **120A** is also driven to the predetermined mass flow rate and the bleed valve **170A** is opened. Optionally, the pressurization profile may be adjusted to account for the bleed of gas from the volume. The bleed valve **170A** is driven to maintain the target pre-flow pressure in the P1 volume **106A**. Upon receipt of a trigger signal, the on/off valve **150A** is opened and the bleed valve **170A** is closed. This allows the apparatus to quickly deliver gas at the predetermined flow rate without requiring exact timing for the turn on time $t_{sub.2}$.

(179) Thus, it is possible to bleed excess gas to maintain the target pre-flow pressure in the P1 volume **106A** while simultaneously maintaining the proportional valve **120A** at the predetermined flow rate. In the embodiments where a fixed characterized restrictor and on/off valve are used, such as the embodiment shown in FIG. **30B**, it is possible to maintain the target pre-flow pressure in the P1 volume **106A** by varying the mass flow rate across the proportional valve **120A**. Although this does not guarantee that the proportional valve **120A** delivers the desired flow rate at the turn on time $t_{sub.2}$ when the turn on time is not known, it ensures that the closure member of the proportional valve **120A** is close to the desired position. The same technique can be applied to a gas flow activation signal including the turn on time $t_{sub.2}$ with similar benefits in reduced transient turn on times. Improved performance can be obtained when the turn on time $t_{sub.2}$ is known, but this is not strictly necessary. Greatly reduced transient turn on times can still be achieved without known turn on times $t_{sub.2}$ as compared with current commercially available devices.

(180) This offers a further improvement because three boundary conditions are met simultaneously. By controlling the mass flow rate across the proportional valve **120A** at the turn on time $t_{sub.2}$, the delivered mass flow rate through the apparatus **100A** will settle to very near the predetermined mass flow rate. Furthermore, the proportional valve **120A** will not need to move significantly to maintain the pressure in the P1 volume at the target pressure.

(181) In further enhancements, the performance of the apparatus may be characterized such that offsets such as the response time of the on/off valve **150A** may be quantified. The on/off valve **150A** may be opened some additional time in advance of the turn on time $t_{sub.2}$ to further improve system response. For instance, if the response time of the on/off valve **150A** is 50 milliseconds, the on/off valve **150A** may be opened at the opening time $t_{sub.2}$ minus 50 milliseconds to ensure that flow begins at exactly turn on time $t_{sub.2}$. The proportional valve **120A** mass flow rate would then be adjusted accordingly, such that the mass flow rate into the P1 volume **106A** equals the predetermined mass flow rate at the time the on/off valve **150A** is opened. Other response times such as the dead time for the PID control loop may also be characterized to optimize system response. The necessary offsets may be incorporated for the control loop or other delays in the apparatus controller **200A**. In some cases, it is conceived that the system response may vary depending on the mass flow rate, so a map of advance opening times may be stored in the memory

280A of the apparatus controller **200A**. The advance opening times may be applied depending on the predetermined flow rate to achieve optimum system response.

(182) In yet other embodiments, the mass flow rate across the proportional valve **120A** may be adjusted to further optimize the resulting mass flow rate and transient turn on time. It is conceivable that the mass flow rate across the proportional valve **120A** may reach the predetermined mass flow rate at some time in between the opening of the on/off valve **150A** and the turn on time $t_{sub.2}$. Thus, the mass flow rate across the proportional valve **120A** may remain constant until the turn on time $t_{sub.2}$. This ensures that the closure member of the proportional valve **120A** is already in position at the turn on time $t_{sub.2}$. Thus, momentum of the closure member of the proportional valve **120A** is minimized. The proportional valve **120A** can respond to needed changes to maintain the delivered flow rate at the predetermined flow rate as fast as possible, regardless of whether the flow should be increased or decreased to maintain the delivered flow rate at the predetermined flow rate.

Section III

(183) The present invention is directed to an apparatus for splitting a flow of process gas into two individual flows of process gas having a known ratio. In some embodiments, the apparatus may also function as a mass flow controller to deliver a known mass flow of gas to a semiconductor or similar process. Semiconductor fabrication is one industry which demands high performance in control of gas flows. As semiconductor fabrication techniques have advanced, customers have recognized the need for splitting a gas flow into two or more separate flows having known ratios. Modern semiconductor processes require that the mass of the gas flow is tightly controlled, the response time minimized, and the ratio between the gas flows is highly accurate. The present invention delivers improved dynamic performance while maintaining exceptional accuracy in the flows delivered. Furthermore, mixtures of a variety of gases may be split with no loss of accuracy in the delivered ratio of flows.

(184) FIG. **35** shows a schematic view of an apparatus for controlling flow **100B**. The flow path of the process gas is indicated by arrows which illustrate the path that the process gas takes through the apparatus **100B**. The apparatus **100B** has a base **102B** comprising an inlet **104B** and an outlet **110B**. A supply of process gas is delivered to the inlet **104B** of the base **102B**. The process gas then flows through a proportional valve **120B** into a P1 volume **106B** within the base **102B**. The proportional valve **120B** meters the mass flow of process gas which passes to the P1 volume **106B**. The proportional valve **120B** is capable of providing proportional control of the process gas such that it need not be fully open or closed, but instead may have intermediate states to permit control of the mass flow rate of process gas.

(185) A P1 pressure transducer **130B** is fluidly coupled to the P1 volume **106B** so that the P1 pressure transducer **130B** can sample the pressure within the P1 volume **106B**. In some embodiments, the P1 pressure transducer **130B** is physically coupled to the base **102B**, while in others it is remotely mounted and coupled via a tube, hose, or another component such as the proportional valve **120B**. Next, the process gas flows out of the P1 volume **106B** into a first on/off valve **150B**. Internal to the first on/off valve **150B** is a valve seat **152B** and a closure member **154B**. When the apparatus **100B** is delivering process gas, the first on/off valve **150B** is in an open state, such that the valve seat **152B** and the closure member **154B** are not in contact. This permits flow of the process gas, and provides a negligible restriction to fluid flow. When the first on/off valve **150B** is in a closed state the closure member **154B** and the valve seat **152B** are biased into contact by the spring **156B**, stopping the flow of process gas through the first on/off valve **150B**.

(186) Downstream of the valve seat **152B**, a first flow restrictor **160B** is introduced into the flow path to meter flow. The first flow restrictor **160B** provides a restriction to fluid flow, or flow impedance. In some embodiments, the flow impedance is known. However, it is also conceived that the flow impedance of the first flow restrictor **160B** may not be known. In yet other embodiments, the first flow restrictor **160B** may be located upstream of the first on/off valve **150B**.

The first flow restrictor **160B** may be selected to have a specific flow impedance so as to achieve a desired range of gas flow rates. The flow impedance does not need to be characterized across the entire range of anticipated gas flows. Instead, it may merely be selected to have an approximate flow impedance.

(187) The first flow restrictor **160B** may be formed as a porous block, a device having small orifices or channels, or any other means of providing a restriction to process gas flow that is consistent and repeatable across a target dynamic operating range of mass flow rates. The first flow restrictor **160B** has a greater resistance to flow than the passages upstream and downstream of the first flow restrictor **160B**. After passing through the flow restrictor **160B**, the process gas exits through the first outlet **110B**. Generally, the first outlet **110B** of the apparatus **100B** is coupled to a first process header, the first process header directing a plurality of process gases to a first applicator in the processing chamber, the processing chamber being the part of the process equipment where articles are processed.

(188) The P1 volume **106B** is also fluidly coupled to a second on/off valve **151B**, the second on/off valve also having a valve seat **152B** and a closure member **154B**, similar to the first on/off valve **150B**. When in the open state, the second on/off valve **151B** allows gas to flow from the P1 volume **106B**, through a second flow restrictor **161B**, to a second outlet **111B**. The second flow restrictor **161B** may have the same or different construction as the first flow restrictor **160B**. Like the first flow restrictor **160B**, the second flow restrictor **161B** may be located either upstream or downstream of the valve seat **152B** of the on/off valve **151B**. In all cases, the second flow restrictor **161B** has a greater resistance to flow than the passages upstream and downstream of the second flow restrictor **161B**. In some embodiments, the second flow restrictor **161B** has the same resistance to flow as the first flow restrictor **160B**. In other embodiments, the second flow restrictor **161B** has a greater or lesser resistance to flow as compared with the first flow restrictor **160B**. The second outlet **111B** is connected to a second process header which directs a plurality of process gases to a second applicator in the processing chamber. In yet other embodiments, the first and second on/off valves **151B**, **152B** may be replaced with a plurality of similar valves. In yet other embodiments, the first and second on/off valves **151B**, **152B** may be substituted for proportional valves.

(189) In some embodiments, the proportional valve **120B**, pressure transducer **130B**, and first and second on/off valves **150B**, **151B** are all directly mounted to the base **102B** while the first and second flow restrictors **160B**, **161B** are mounted to the first and second on/off valves **150B**, **151B**. Thus, the first and second flow restrictors **160B**, **161B** are indirectly coupled to the base while the proportional valve **120B**, pressure transducer **130B**, and first and second on/off valves **150B**, **151B** are all directly coupled to the base. In yet other embodiments, each of the proportional valve **120B**, pressure transducer **130B**, first and second on/off valves **150B**, **151B**, and first and second flow restrictors **160B**, **161B** are directly coupled to the base **102B**. In yet other embodiments, it is possible to remotely mount one or more of the proportional valve **120B**, pressure transducer **130B**, first and second on/off valves **150B**, **151B**, and first and second flow restrictors **160B**, **161B** such that they are operably coupled but neither directly nor indirectly mounted to the base **102B**. In yet other embodiments, the base **102B** may be omitted.

(190) Optionally, the apparatus **100B** comprises one or more P2 pressure transducers downstream of the flow restrictors **160B**, **161B** and the on/off valves **150B**, **151B**. The P2 pressure transducer is used to measure the pressure differential across the flow restrictors **160B**, **161B**. In some embodiments, the pressure downstream of only one of the two flow restrictors **160B**, **161B** is measured. In other embodiments, the pressure downstream of both of the flow restrictors **160B**, **161B** is measured. In yet other embodiments, the P2 pressure downstream of either or both flow restrictors **160B**, **161B** may be obtained from another apparatus **100B** connected to the first or second process header, with the readings communicated to the first apparatus **100B**.

(191) Optionally, temperature sensors may be employed to further enhance the accuracy of the

apparatus **100B**. A temperature sensor **114B** is shown in FIG. **35**, located within the base **102B** so that it can measure the temperature near the P1 volume **106B**. Additional temperature sensors **114B** may be employed in a variety of locations, including the proportional valve **120B**, the pressure transducer **130B**, and the first and second on/off valves **150B**, **151B**.

(192) In yet a further embodiment, the apparatus **100B** may omit the proportional valve **120B** and the P1 pressure transducer. In this embodiment, the process gas is supplied to the gas inlet **104B** at a known pressure. The pressure of the process gas may be varied external to the apparatus **100B** and may change over time.

(193) Turning to FIG. **36**, a block diagram illustrates the control system for the apparatus **100B** of FIG. **35**. This block diagram shows a controller **350B** which comprises an apparatus controller **200B** and a system controller **300B**. The apparatus controller **200B** provides all control functions for the apparatus **100B**. The apparatus controller **200B** has a communication interface **210B**, a proportional valve controller **220B**, a pressure transducer interface **230B**, an on/off valve controller **250B**, a temperature sensor interface **260B**, a processor **270B**, and memory **280B**. The communication interface **210B** is configured to provide a communications link between the apparatus controller **200B** and the system controller **300B**. Optionally, the temperature sensor interface **260B** may be omitted if the additional accuracy provided by the temperature sensor **114B** is not required.

(194) The system controller **300B** has a corresponding communication interface **310B**, a processor **370B**, and memory **380B**. The system controller **300B** coordinates all high-level functions necessary to perform the desired process. The communication interface **310B** of the system controller **300B** sends and receives commands through a communications bus **390B**. The communications bus **390B** connects to the communication interface **210B** of the apparatus controller **200B** of the apparatus **100B**. The communications bus **390B** may connect to a single apparatus controller **200B**, or it may connect to a plurality of apparatus controllers **200B**, each apparatus controller **200B** operating a distinct device. Not all apparatus controllers **200B** need control an apparatus for controlling gas flow **100B**. Instead, other types of process equipment may also be controlled. Furthermore, there may be a plurality of communications buses **390B** to connect all the devices required to perform the desired process.

(195) The apparatus **100B** according to the present invention is capable of serving several roles in a system for processing articles. In one configuration, the apparatus **100B** is used with a process gas having a known composition. In this configuration, either one of the first and second on/off valves **150B**, **151B** may be operated to deliver a known mass flow rate of the process gas to either one of the first or second process headers. In this way, a single apparatus **100B** can serve as a mass flow controller for two separate applicators. Full control over a wide range of mass flow rates can be achieved for both applicators. However, this configuration requires that the memory **280B** of the apparatus controller **200B** store a gas map containing information permitting precise calculation of the pressure drop across one or more of the flow restrictors **160B**, **161B** required to achieve a predetermined mass flow rate at the respective outlet **110B**, **111B**. The gas map must be calibrated for the process gas utilized and corresponds to the flow impedance of the flow restrictors **160B**, **161B** across the operating range of the apparatus **100B**.

(196) In a second configuration, the apparatus **100B** is used to simultaneously deliver process gas at a known ratio to the outlets **110B**, **111B**. The ratio is determined by the ratio of the flow impedances of the first and second flow restrictors **160B**, **161B**. Thus, it is possible to simultaneously deliver two flows of process gas to the first and second outlets **110B**, **111B** at a fixed ratio. In this configuration, the composition of the process gas does not need to be known, and may vary with time. This is because the ratio of the flow rates is determined by the ratio of the resistances to flow of the flow restrictors **160B**, **161B** and remains constant regardless of the gas mixture. There is no need to store a gas map for the process gas in the memory **280B** of the apparatus controller **200B** in this configuration.

(197) In the second configuration, it is also possible to accurately control a mass flow rate of one of the two flows delivered to the outlets **110B**, **111B** while simultaneously providing a known ratio between the flow flows. This requires a known composition of gas and an appropriately calibrated gas map, but permits the user to split the flow of process gas into two separate flows while simultaneously controlling the mass flow rate at one of the two outlets.

(198) By controlling the ratio between the flow impedances for each of the flow restrictors **160B**, **161B**, an accuracy of 99% or greater can be achieved for the splitting of the flow of process gas, even where the mixture of process gas is unknown. In some embodiments, it is also possible to extend the apparatus **100B** to include more than two on/off valves **150B**, **151B** to split the process gas into three or more different flows of known ratios. It is also possible to provide two or more proportional valves **120B** in the same device, with each proportional valve **120B** directly connected to a single on/off valve, enabling separate control of the pressure differential across each of the flow restrictors **160B**, **161B**. This enables variable control over the flow ratio between the outlets.

(199) The present invention also minimizes flow spikes and reduces the transient turn-on and turn-off times for the flows of process gas. By locating the flow restrictors **160B**, **161B** in the respective on/off valves **150B**, **151B**, the volume that must be charged or bled during turn-on and turn-off is minimized. The time required to start and stop flow is largely driven by the speed at which the on/off valves **150B**, **151B** can open and close, and pressure and flow pulses are also minimized. Typical turn-on times for the on/off valves **150B**, **151B** are between 50 and 100 milliseconds, permitting exceptionally fast control over the delivery of process gas to the respective applicators in the processing chamber. Furthermore, the pressure of the process gas supplied to the apparatus **100B** is unimportant because the ratio between the mass flow rates remains constant regardless of the supplied pressure.

(200) In yet another embodiment shown schematically in FIG. **38**, an apparatus **400B** has a second proportional valve **121B** and a second pressure transducer **131B** are incorporated in the gas flow path between the P1 pressure transducer and the first on/off valve **150B**. A second pressure transducer **131B** monitors the pressure between the proportional valve **121B** and the flow restrictor **160B**. This variation enables independent control of gas flows delivered to both the first and second applicators. It is possible to deliver gas flows which deviate from the ratio of the flow impedance between the first and second flow restrictors **160B**, **161B** by separately controlling the pressure of the gas in the P1 volume **106B** and the pressure between the proportional valve **121B** and the first flow restrictor **160B**. In some embodiments, the second pressure transducer **131B** may be omitted.

(201) In the embodiment shown schematically in FIG. **39**, three separate gas outlets are shown in an apparatus **500B**. The three gas outlets may be directed to first, second, and third applicators, or may be combined or divided further downstream as desired for a particular process. The apparatus substitutes the first on/off valve **150B** for a second proportional valve **121B**. This allows three separate gas flows to be delivered at known ratios determined by the combination of the first, second, and third flow restrictors **160B**, **161B**, **162B** when the proportional valve **121B**, the second on/off valve **151B**, and the third on/off valve **152B** are all in the fully open state. Furthermore, the valves **121B**, **151B**, **152B** may be selectively closed to deliver specific mass flows to one gas outlet at a time, or the gas may be split into two flows by closing only one of the three valves **121B**, **151B**, **152B**. Furthermore, the substitution of the proportional valve **121B** for the first on/off valve **150B** enables further control of the gas flow delivered to the first gas outlet. This arrangement is similar to that of FIG. **38**. It will be noted that the second pressure transducer **131B** of FIG. **38** is not required in all embodiments.

(202) In the embodiment of FIG. **40**, an apparatus **600B** has a proportional valve **121B** in lieu of a first on/off valve. This enables identical function to the apparatus **100B** with the added capability of further varying the gas flow rates through the gas outlets as discussed above. Nearly total control over the flow rates can be achieved with this configuration without the need for two separate mass flow controllers. Furthermore, approximate flow splitting can be achieved even when the exact

composition of the gas is not known.

(203) In a first method of operating the apparatuses, a process gas having a known composition is supplied to the inlet **104B**. The system controller **300B** transmits a command to the apparatus controller **200B** to flow gas from the first outlet **110B** at a predetermined mass flow rate. In response, the proportional valve **120B** is opened to permit the process gas to flow into the P1 volume **106B**. The first on/off valve **150B** is opened while the second on/off valve **151B** is closed. The apparatus controller **200B** causes the proportional valve **120B** to adjust its position to adjust the pressure in the P1 volume **106B** based on feedback from the P1 pressure transducer **130B**. This causes a known mass flow rate to be delivered through the first on/off valve **150B** and the first flow restrictor **160B**, and out of the first outlet **110B**. Subsequently, the apparatus controller **200B** receives a command to flow gas through the second outlet **111B** at a predetermined mass flow rate. The first on/off valve **150B** is closed and the second on/off valve **151B** is opened. The proportional valve **120B** adjusts its position to drive the pressure in the P1 volume **106B** to the pressure required to deliver the predetermined flow rate. In this method, a gas map which is accurately calibrated to the process gas being used is loaded into the memory **280B** of the apparatus controller **200B**, which ensures that the correct mass flow rate is delivered to the appropriate outlet. Furthermore, the flow impedance for each of the flow restrictors **160B**, **161B** is known with a high degree of accuracy. In some embodiments, there may be one gas map for each of the flow restrictors **160B**, **161B** to ensure even greater accuracy.

(204) In a second method, a process gas having known or unknown composition is supplied to the inlet **104B**. The system controller **300B** transmits a command to flow gas to both of the outlets **110B**, **111B**. In response, the proportional valve **120B** and the on/off valves **150B**, **151B** open and the process gas flows out of both outlets **110B**, **111B**. The flow rate may be adjusted by altering the position of the proportional valve **120B**. In the event that the process gas is known, one of the outlets **110B**, **111B** may be driven to a specific mass flow rate based on a calibrated gas map. In the event that the process gas is unknown, an estimated gas map may be used to drive the mass flow rate through one of the outlets **110B**, **111B**. The flows from the outlets **110B**, **111B** may be halted by closing one or more of the on/off valves **150B**, **151B**.

(205) In a third method, a process gas of known or unknown composition is supplied to the inlet **104B**. The system controller **300B** transmits a command to transition the first on/off valve **150B** to the fully open state and the second on/off valve **151B** to the fully closed state. A controlled flow of process gas is delivered to the first gas outlet **110B** and no gas is delivered to the second gas outlet **111B**. Subsequently, the system controller **300B** transmits a command to transition the first on/off valve **150B** to the fully closed state and the second on/off valve **151B** to the fully open state. This results in the delivery of a controlled flow of process gas to the second gas outlet **111B** while no gas is delivered to the first gas outlet **110B**. Subsequently, the system controller **300B** transmits a command to transition both the first and second on/off valves **150B**, **151B** to the fully open state. First and second controlled flows of process gas are delivered to the first and second gas outlets **110B**, **111B**. In other embodiments of this method, the sequence is altered such that the second on/off valve **151B** is opened before the first on/off valve **150B** or the first or second on/off valves **150B**, **151B** are closed to transition from delivering process gas to the first and second gas outlets **110B**, **111B** to delivering process gas to only one of the first and second gas outlets **110B**, **111B**. In yet other embodiments, a proportional valve **121B** may be substituted for one or both of the first and second on/off valves **150B**, **151B** and may be operated at a state intermediate between the fully open and fully closed states. In yet other embodiments, a second proportional valve **121B** may be introduced upstream of either one of the first and second flow restrictors **160B**, **161B**.

Section IV

(206) The present invention is directed to a laminar flow restrictor for use in an apparatus for controlling gas flow. In some embodiments, the apparatus may function as a mass flow controller to deliver a known mass flow of gas to a semiconductor or similar process. Semiconductor

fabrication is one industry which demands high performance in control of gas flows. As semiconductor fabrication techniques have advanced, customers have recognized the need for flow control devices with increased accuracy and repeatability in the mass of the delivered gas flows. Modern semiconductor processes require that the mass of the gas flow is tightly controlled, the response time minimized, and the gas flow is highly accurate. The present invention delivers improved accuracy and repeatability in the delivered flows.

(207) FIG. 41 shows a schematic of an exemplary processing system 1000C utilizing one or more laminar flow restrictors. The processing system 1000C may utilize a plurality of apparatus for controlling flow 100C fluidly coupled to a processing chamber 1300C. The plurality of apparatus for controlling flow 100C are used to supply one or more different process gases to the processing chamber 1300C. Articles such as semiconductors may be processed within the processing chamber 1300C. Valves 1100C isolate each of the apparatus for controlling flow 100C from the processing chamber 1300C, enabling each of the apparatus for controlling flow 100C to be selectively connected or isolated from the processing chamber 1300C, facilitating a wide variety of different processing steps. The processing chamber 1300C may contain an applicator to apply process gases delivered by the plurality of apparatus for controlling flow 100C, enabling selective or diffuse distribution of the gas supplied by the plurality of apparatus for controlling flow 100C. In addition, the processing system 1000C may further comprise a vacuum source 1200C which is isolated from the processing chamber 1300C by a valve 1100C to enable evacuation of process gases or facilitate purging one or more of the apparatus for controlling flow 100C to enable switching between process gases in the same apparatus for controlling flow 100C. Optionally, the apparatus for controlling flow 100C may be mass flow controllers, flow splitters, or any other device which controls the flow of a process gas in a processing system. Furthermore, the valves 1100C may be integrated into the apparatus for controlling flow 100C if so desired.

(208) Processes that may be performed in the processing system 1000C may include wet cleaning, photolithography, ion implantation, dry etching, atomic layer etching, wet etching, plasma ashing, rapid thermal annealing, furnace annealing, thermal oxidation, chemical vapor deposition, atomic layer deposition, physical vapor deposition, molecular beam epitaxy, laser lift-off, electrochemical deposition, chemical-mechanical polishing, wafer testing, or any other process utilizing controlled volumes of a process gas.

(209) FIG. 42 shows a schematic of an exemplary mass flow controller 101C, which is one type of apparatus for controlling flow 100C that may be utilized in the processing system 1000C. The mass flow controller 101C has a gas supply of a process gas fluidly coupled to an inlet 104C. The inlet is fluidly coupled to a proportional valve 120C which is capable of varying the volume of process gas flowing through the proportional valve 120C. The proportional valve 120C meters the mass flow of process gas which passes to the P1 volume 106C. The proportional valve 120C is capable of providing proportional control of the process gas such that it need not be fully open or closed, but instead may have intermediate states to permit control of the mass flow rate of process gas.

(210) A P1 volume 106C is fluidly coupled to the proportional valve 120C, the P1 volume 106C being the sum of all the volume within the mass flow controller 101C between the proportional valve 120C and a flow restrictor 160C. A pressure transducer 130C is fluidly coupled to the P1 volume 106C to enable measurement of the pressure within the P1 volume 106C. An on/off valve 150C is located between the flow restrictor 160C and the proportional valve 120C and may be used to completely halt flow of the process gas out of the P1 volume 106C. Optionally, the flow restrictor 160C may be located between the on/off valve 150C and the proportional valve 120C in an alternate configuration. Finally, the flow restrictor 160C is fluidly coupled to an outlet 110C of the mass flow controller 101C. In the processing system, the outlet 110C is fluidly coupled to a valve 1100C or directly to the processing chamber 1300C.

(211) Internal to the first on/off valve 150C is a valve seat and a closure member. When the apparatus 100C is delivering process gas, the first on/off valve 150C is in an open state, such that

the valve seat and the closure member are not in contact. This permits flow of the process gas, and provides a negligible restriction to fluid flow. When the first on/off valve **150C** is in a closed state the closure member and the valve seat are biased into contact by a spring, stopping the flow of process gas through the first on/off valve **150C**.

(212) The flow restrictor **160C** is used, in combination with the proportional valve **120C**, to meter flow of the process gas. In most embodiments, the flow restrictor **160C** provides a known restriction to fluid flow. The first characterized flow restrictor **160C** may be selected to have a specific flow impedance so as to deliver a desired range of mass flow rates of a given process gas. The flow restrictor **160C** has a greater resistance to flow than the passages upstream and downstream of the flow restrictor **160C**.

(213) Optionally, the mass flow controller **101C** comprises one or more P2 pressure transducers downstream of the flow restrictor **160C** and the on/off valve **150C**. The P2 pressure transducer is used to measure the pressure differential across the flow restrictor **160C**. In some embodiments, the P2 pressure downstream of the flow restrictor **160C** may be obtained from another apparatus **100C** connected to the processing chamber, with the readings communicated to the mass flow controller **101C**.

(214) Optionally, temperature sensors may be employed to further enhance the accuracy of the mass flow controller **101C**. They may be mounted in the base of the mass flow controller **101C** near the P1 volume **106C**. Additional temperature sensors may be employed in a variety of locations, including the proportional valve **120C**, the pressure transducer **130C**, and the on/off valve **150C**.

(215) Turning to FIGS. **43-49**, a first embodiment of the flow restrictor **160C** is shown in greater detail. The flow restrictor **160C** is constructed as a plurality of layers forming a restrictor stack **170C**. The restrictor stack **170C** may take the form of an elongated rectangular shape as shown in FIG. **43**. The flow restrictor **160C** extends from a first end **161C** to a second end **162C** along a longitudinal axis A-A. A plurality of layers **210C** comprising flow passages are sandwiched between a plurality of outer layers **220C** which do not comprise flow passages. The flow restrictor **160C** has a first side **163C** formed of the pluralities of layers **210C**, **220C** and an opposite second side **164C**. The flow restrictor **160C** further comprises a front face **165C** and an opposite rear face **166C**. The outer layers **220C** enclose the flow passages on opposite sides of the layers **210C** comprising flow passages. The outer layers **220C** may or may not have the same thickness as the layers **210C** comprising flow passages. A selection of the layers **210C** is shown in FIG. **44**, which illustrates portions of the flow passages and the configuration of the layers **210C**. Each of the layers **210C** extend from a first end **213C** to a second end **214C**. Portions of the plurality of flow passages can be seen in FIG. **44**. The details of the flow passages will be discussed in greater detail below.

(216) Turning to FIGS. **45A** and **45B**, the layers **210C** comprise a plurality of apertures **212C** formed at opposite ends **213C**, **214C** of the layers **210C**. This enables gas to flow along the layers **210C** from the first end **213C** to the second end **214C** along the longitudinal axis A-A. In alternate embodiments, the apertures **212C** need not be on opposite ends and may instead be formed on adjacent sides or may be formed exclusively on a single end. The apertures **212C** may also be formed so that gas flows across the shorter direction of the rectangular layers **210C**, perpendicular to the longitudinal axis A-A. The layers **210C** also need not be rectangular and may be square or any other desired shape. It is further contemplated that an aperture may be formed into the plane of the layers **210C**, permitting gas to flow perpendicular to the planes of the layers **210C**, then turn a corner and flow in the plane of the layers **210C**. The specific arrangement of the apertures **212C**, the shape of the layers **210C**, and the shape of the resulting flow restrictor **160C** may be adapted as desired depending on the shape of the flow passage which receives the resulting flow restrictor **160C**. It is even contemplated that the flow restrictor **160C** may have an annular configuration, with apertures **212C** formed into a circumference of the flow restrictor **160C** and/or apertures **212C** formed so that gas flows perpendicular to the planes of some or all of the layers **210C**.

(217) FIG. 46 shows an exploded view of the layers 210C. The layers 210C comprise two first layers 230C and two second layers 260C. As best seen in FIGS. 48 and 49, the first layer 230C has a first side 231C, a second side 232C, a third side 233C, a fourth side 234C, a front face 235C, and an opposite rear face 236C. The second layer 260C has a first side 261C, a second side 262C, a third side 263C, a fourth side 264C, a front face 265C, and an opposite rear face 266C. The first layer 230C has a series of flow passages comprising entry passages 237C, U passages 238C, and longitudinal passages 239C. The entry passages and the U passages are each only formed into a portion of the thickness of the first layer 230C while the longitudinal passages 239C extend through the entirety of the thickness of the first layer 230C. The second layer 260C also has entry passages 267C and U passages 268C formed into the front face 265C that correspond to the entry passages 237C and U passages 238C of the first layer 240C. When a first layer 230C and a second layer 260C are stacked with the front faces 235C, 265C facing one another, the entry passages 237C, 267C form apertures 212C on the first and second sides 231C, 261C, 232C, 262C of the layers 230C, 260C. As is best shown in FIG. 47, in combination with additional first layers 230C and second layers 260C, a plurality of flow passages 270C are formed, extending from apertures 212C on one end 213C of the plurality of layers 210C to the opposite second end 214C of the plurality of layers 210C.

(218) Returning to FIG. 45A, the apertures 212C have a first edge 215C, a second edge 216C, a third edge 217C, and a fourth edge 218C. The first edge 215C is formed by the first layer 230C, the second edge 216C is formed by the second layer 260C, and the third and fourth edges 217C, 218C are each formed by a portion of the first layer 230C and a portion of the second layer 260C.

(219) The flow passages 270C may be varied in any desired manner to achieve a desired flow impedance. For instance, the number of flow passages 270C may be increased or decreased by reducing or increasing the number of the plurality of layers 210C. Furthermore, the length of the flow passages 270C may be increased or decreased by changing the number of times that the flow passages 270C double back on themselves, changing the resulting number of U passages 238C, 268C and longitudinal passages 239C. A greater or fewer number of flow passages 270C may be formed into pairs of first and second layers 230C, 260C. The width of the flow passages 270C may also be increased or decreased, and the thickness of the first and second layers 230C, 260C may be varied. Indeed, it is not necessary that the same thickness be used for every pair of first and second layers 230C, 260C. Each layer within the plurality of layers 210C could be individually varied to alter the resulting flow impedance of the flow restrictor 160C.

(220) The flow restrictor 160C is manufactured by first etching each of the layers 210C individually or in an array. The layers 210C may all be formed of the same material or may be formed of different materials. The etching may be carried out in a single step or in a series of steps to achieve the multiple depths required. Alternative processes such as laser ablation, micromachining, or other known processes may also be used. Once the plurality of layers 210C have been formed, they are assembled with the non-etched outer layers 220C and joined by diffusion bonding. Again, alternative techniques such as conventional bonding with adhesives, welding, or similar processes may also be used as is known in the art. The resulting stack of layers 210C, 220C is joined, sealing the flow passages 270C and forming the flow restrictor 160C. Subsequent finishing steps can be performed to alter the overall shape or size of the flow restrictor 160C to suit the dimensions of the flow passages into which the flow restrictor 160C is installed. These processes may include grinding, machining, laser cutting, water jetting, or other known techniques. Indeed, the flow restrictor 160C does not need to remain rectangular and may be formed into cylindrical shapes as will be discussed further below.

(221) Turning to FIGS. 50-56, a second embodiment of the flow restrictor 300C is best shown in FIG. 50. Where not explicitly noted, the reference numerals are identical to those of the first embodiment of the flow restrictor 160C. The second embodiment of the flow restrictor 300C extends from a first end 302C to a second end 303C along a longitudinal axis A-A and is also

formed of a plurality of layers **310C** having flow passages and a plurality of outer layers **320C** which do not have flow passages. After bonding, the layers **310C**, **320C** are post-processed into a cylindrical shape which facilitates insertion into a cylindrical bore, enabling easy installation of the flow restrictor **300C** into a valve or other flow device.

(222) As shown in FIG. **51**, a selection of the layers **310C** are shown in perspective. The layers **310C** extend from a first end **313C** to a second end **314C** opposite the first end **313C**. FIGS. **52A** and **52B** best illustrate the apertures **312C** formed into the first end **313C** of the layers **310C**. As can also be seen, the layers **310C** comprise two first layers **330C** and two second layers **360C**. As best seen in FIG. **52B**, the apertures **312C** have a first edge **315C**, a second edge **316C** opposite the first edge **315C**, a third edge **317C**, and a fourth edge **318C** opposite the third edge **317C**. The first and second edges **315C**, **316C** are formed by the first layers **330C**. The third and fourth edges **317C**, **318C** are each formed by the second layer **360C**.

(223) An exploded view of the layers **310C** is shown in FIG. **53**, better illustrating the flow passages of the restrictor **300C**. FIGS. **55** and **56** illustrate the first layer **330C** and the second layer **360C**, respectively. The first layer **330C** has a first side **331C**, a second side **332C**, a third side **333C**, a fourth side **334C**, a front face **335C**, and an opposite rear face **336C**. The second layer **360C** has a first side **361C**, a second side **362C**, a third side **363C**, a fourth side **364C**, a front face **365C**, and an opposite rear face **366C**. The first layer **330C** has a series of longitudinal passages **339C** that terminate in layer transition apertures **340C**. The longitudinal passages **339C** and the layer transition apertures **340C** extend through the entirety of the first layer **330C**. The second layer **360C** has notches **369C** that extend from the first and second sides **361C**, **362C**. The notches **369C** also extend through the entirety of the second layer **360C**. As can be best seen in FIG. **54**, the apertures **312C** are formed by the open ends of the notches **369C** when the layers **330C**, **360C** are alternately stacked as shown. Flow passages **370C** are formed by the stacking of the layers **330C**, **360C** as shown. In alternate embodiments, the layer transition apertures may be formed in a variety of shapes and may be formed with or without flow passage contouring at the ends of the channel, or with contouring of different shapes.

(224) Once again, a plurality of the layers **330C**, **360C** are stacked and assembled with the outer layers **320C**. The layers are then bonded through diffusion bonding or a similar technique. The resulting restrictor stack is then ground or machined into a cylindrical shape as shown in FIG. **50**. This cylindrical shape also incorporates annular grooves which facilitate the mounting of a seal which seals the flow restrictor **300C** into a bore of a device to ensure that the only gas passing by the flow restrictor **300C** must pass through the passages **370C**. In other embodiments, the final part may be machined into different shapes, or alternatively left in its raw shape formed by the bonded restrictor stack.

(225) A third embodiment of the flow restrictor **400C** is shown in FIGS. **57-62**. FIG. **57** shows a selection of the plurality of layers **410C** forming the flow passages of the flow restrictor **400C**. The outer layers are not shown in this embodiment as they are substantially identical to those of the other embodiments. The plurality of layers **410C** extend from a first end **413C** to a second end **414C** opposite the first end **413C**. As best shown in FIG. **58**, apertures **412C** are formed in the first end **413C** and the second end **414C** to permit passage of gas into and out of the flow restrictor **400C**. FIG. **59** shows an exploded view of the plurality of layers **410C** to better illustrate the flow passages. As can be seen, the plurality of layers **410C** comprise two first layers **430C** and two second layers **460C**.

(226) FIGS. **61** and **62** illustrate the first layer **430C** and the second layer **460C**, respectively. The first layer **430C** has a first side **431C**, a second side **432C**, a third side **433C**, a fourth side **434C**, a front face **435C**, and an opposite rear face **436C**. The second layer **460C** has a first side **461C**, a second side **462C**, a third side **463C**, a fourth side **464C**, a front face **465C**, and an opposite rear face **466C**. The first layer **430C** has a series of longitudinal passages **439C** having an elongated configuration with straight sides and a radius at each end. The second layer **460C** has notches **469C**

that transition from a u-shape having parallel sides to angled sides which increase in width as they approach the first side **461C** or second side **462C** of the second layer **460C**. The notches **469C** overlap with the longitudinal passages **439C** when the first and second layers are aligned. The second layer **460C** also has D-shaped apertures **468C** which allow the connection of two adjacent longitudinal passages **439C** to increase the effective length of the flow passage from one aperture **412C** to another aperture **412C**. There is no limit to the number of D-shaped apertures **468C** which may be employed. Furthermore, there is no need to limit the apertures **468C** to a D shape, and they may be any desired shape to facilitate a connection between adjacent longitudinal passages **439C**. In alternate embodiments the notches **469C** can be shaped differently. For instance, shapes such as rectangular, wedge, or other shapes may be used. Additionally, longitudinal passages **439C** can have contouring in them to improve flow characteristics. Thus, the longitudinal passages **439C** need not be formed with a constant width, and may have varying widths at either ends or anywhere along their length. In yet further embodiments a third layer (or a plurality of layers) can be interleaved between the first layer **430C** and the second layer **460C** such that each first layer **430C** only contacts one second layer **460C**, and the apertures **468C** between subsequent sheets do not allow flow transitions except for adjacent first and second layers **430C**, **460C**.

(227) As can be best seen in FIG. **60**, the apertures **412C** are formed by the open ends of the notches **469C** when the layers **430C**, **460C** are alternately stacked as shown. Flow passages **470C** are formed by the stacking of the layers **430C**, **460C** as shown. The layers **430C**, **460C** are of equal thickness in this embodiment, but may have different thicknesses if desired.

(228) A fourth embodiment of the flow restrictor **500C** is shown in FIGS. **63-69**. FIG. **63** shows a selection of the plurality of layers **510C** forming the flow passages of the flow restrictor **500C**. The outer layers are not shown in this embodiment as they are substantially identical to those of the other embodiments. The plurality of layers **510C** extend from a first end **513C** to a second end **514C** opposite the first end **513C**. As best shown in FIG. **64**, apertures **512C** are formed in the first end **513C** and the second end **514C** to permit passage of gas into and out of the flow restrictor **500C**. FIG. **65** shows an exploded view of the plurality of layers **510C** to better illustrate the flow passages. As can be seen, the plurality of layers **510C** comprise a first layer **530C**, a second layer **560C**, and a third layer **580C**.

(229) FIGS. **67-69** illustrate the first layer **530C**, the second layer **560C**, and the third layer **580C**, respectively. The first layer **530C** has a first side **531C**, a second side **532C**, a third side **533C**, a fourth side **534C**, a front face **535C**, and an opposite rear face **536C**. The second layer **560C** has a first side **561C**, a second side **562C**, a third side **563C**, a fourth side **564C**, a front face **565C**, and an opposite rear face **566C**. The first layer **530C** has a series of longitudinal passages **539C** having an elongate configuration with straight sides and a radius at each end. The second layer **560C** has notches **569C** that transition from a u-shape having parallel sides to angled sides which increase in width as they approach the first side **561C** or second side **562C** of the second layer **560C**. The notches **569C** overlap with the longitudinal passages **539C** when the first and second layers are aligned. The third layer **580C** has a first side **581C**, a second side **582C**, a third side **583C**, a fourth side **584C**, a front face **585C**, and an opposite rear face **586C**. As can be best seen in FIG. **66**, the apertures **512C** are formed by the open ends of the notches **569C** when the layers **530C**, **560C** are alternately stacked as shown. Flow passages **570C** are formed by the stacking of the layers **530C**, **560C** as shown. The layers **530C**, **560C** are of equal thickness in this embodiment, but may have different thicknesses if desired. The third layer may be useful for decreasing the density of the flow passages, ensuring that flow is more evenly distributed across the cross-sectional area of the flow restrictor **500C**. This is particularly useful for producing very high flow impedance flow restrictors.

(230) A fifth embodiment of the flow restrictor **600C** is shown in FIGS. **70-76**. FIG. **70** shows the flow restrictor **600C** in perspective. The flow restrictor **600C** extends from a first end **602C** to a second end **603C** and has outer layers **620C** which surround layers **610C** which have flow passages therein. A selection of the layers **610C** are shown in FIG. **71** in perspective view. These layers **610C**

extend from a first end **613C** to a second end **614C**, with apertures **612C** on the first and second ends **613C**, **614C**. An exploded view of the layers **610C** is shown in FIG. 73, illustrating two first layers **630C** and two second layers **660C**.

(231) The first layer **630C** and the second layer **660C** are illustrated in FIGS. 75 and 76. The first layer **630C** has a first side **631C**, a second side **632C**, a third side **633C**, a fourth side **634C**, a front face **635C**, and an opposite rear face **636C**. The second layer **660C** has a first side **661C**, a second side **662C**, a third side **663C**, a fourth side **664C**, a front face **665C**, and an opposite rear face **666C**. The first layer **630C** has a series of longitudinal passages **639C** having an elongated configuration which meet with U shaped portions **640C** or with openings **641C**. The second layer **660C** is free of any flow passages or other features. As can be seen, in the flow restrictor **600C**, gas remains exclusively on a single layer **630C** and does not transition between first and second layers **630C**, **660C**. Instead, it enters through an opening **641C** at the first side **631C**, travels down a longitudinal passage **639C**, returns along a U shaped portion **640C** at least two times, then exits through an opening **641C** on the second side **632C**. The exact flow path may be altered to zig-zag, utilize more than two U shaped portions **640C**, no U shaped portions **640C**, or take any other path on the layer **630C**. However, it never flows through the second layer **660C** in this embodiment. The longitudinal passages **639C**, U shaped portions **640C**, and openings **641C** all extend through the entirety of the thickness of the first layer **630C**. In alternate configurations, single sheet flow may be obtained by forming the flow passage depth only partially through the sheet such that the sheet dimensions remain intact during assembly prior to bonding.

(232) As best shown in FIG. 74, flow passages **670C** are formed by the stacking of the layers **630C**, **660C** as shown. The layers **630C**, **660C** are of equal thickness in this embodiment, but may have different thicknesses if desired. The layers **630C**, **660C** are formed individually of different materials having a different reactivity when subjected to etching chemicals. The material of the first layer **630C** may be more reactive than the material of the second layer **660C**, facilitating effective etching of the first layer **630C** without significant etching of the second layer **660C**. Layer pairs are formed by assembling one first layer **630C** with one second layer **660C**. The layer pairs are then diffusion bonded so they cannot be readily separated. As discussed above, other bonding techniques may be utilized. Then, the layer pairs are etched so that the flow passages **670C** are formed into the first layer **630C** without etching the second layer **660C**. The layer pairs are then assembled into the plurality of layers **610C** having flow passages **670C**. Outer layers **620C** are also assembled with the plurality of layers **610C** having the flow passages **670C**. Finally, the layers **610C**, **620C** are diffusion bonded together. Optionally, post processing such as grinding may be used to form the flow restrictor **600C** and adapt it for installation into a flow passage of a device.

(233) It should be noted that the flow passages do not need to extend straight from one end of the flow restrictor to the other end of the flow restrictor, or double back in parallel rows. Instead, it is conceived that the flow passages may zig-zag, arc, or take any other path needed to achieve the desired flow impedance in the completed flow restrictor. Multiple layer transitions may also be made, enabling the use of flow passages which fork and rejoin, transition across more than two or three layers, or the like. It is further conceived that flow restrictors may incorporate features of specific embodiments in combination, such that a hybrid of the disclosed embodiments may be constructed. The above-disclosed restrictor designs can be used to achieve highly laminar flow and high part to part reproducibility. This high reproducibility reduces calibration requirements when manufacturing flow control devices utilizing one or more laminar flow elements.

(234) Details illustrating a method of forming the flow restrictors according to the present invention are shown in FIGS. 77-81. FIG. 77 shows a plurality of layer blanks **710C** in an exploded view. Each of the layer blanks has a first edge **711C**, a second edge **712C** opposite the first edge, a third edge **713C**, and a fourth edge **714C** opposite the third edge. The layer blanks **710C** further comprise a front face **715C** and a rear face **716C** opposite the front face **715C**. The layer blanks **710C** are formed into first layers **730C** and second layers **760C** as further illustrated in FIGS. 78

and 79. The first layer 730C is modified from a layer blank 710C by forming a second cavity 732C into the first layer 730C. The second layer 760C is modified from a layer blank 710C by forming a first cavity 761C and a third cavity 763C into the second layer 760C. The first, second, and third cavities 761C, 732C, 763C are formed into the front faces 715C of their respective first and second layers 730C, 760C. Preferably the cavities 761C, 732C, 763C are formed through the thickness of the layers 730C, 760C. In some embodiments, some or all of the cavities 761C, 732C, 763C may be formed only partially through the thickness of the layers 730C, 760C. In the illustrated method, the cavities 761C, 732C, 763C are formed from the front face 715C to the rear face 716C. The cavities 761C, 732C, 763C are spaced from the first, second, third, and fourth edges 711C, 712C, 713C, 714C of the layer blanks 710C.

(235) The cavities 761C, 732C, 763C are formed by etching the layer blanks 710C. Alternate processes are available such as micromachining, laser ablation, or other known techniques. As illustrated in FIG. 80, a restrictor stack 770C is formed from the plurality of layers 730C, 760C. Subsequent to formation of the cavities 761C, 732C, 763C, the layers 730C, 760C are stacked in alternating layers, ensuring that the layers 730C, 760C are kept in alignment so that the second cavity 762C overlaps with the first and third cavities 761C, 763C. The layers 730C, 760C are then bonded to form the restrictor stack 770C as a unitary component. The layers 730C, 760C may be bonded by diffusion bonding, welding, gluing, or any other known technique.

(236) The restrictor stack 770C comprises a first unfinished end 771C formed by the first edges 711C of the first and second layers 730C, 760C. An opposite second unfinished end 772C is formed by the second edges 712C of the first and second layers 730C, 760C of the restrictor stack 770C. As can be seen, no cavities are exposed on the unfinished ends 771C, 772C. In alternate embodiments, only one of the layers 730C, 760C need have cavities, with the other layers 730C, 760C being free of cavities. This allows formation of restrictor such as those shown in FIGS. 70-76. In yet other embodiments, three or more different types of layers may be utilized such as is shown in FIGS. 63-67. The layers need not be alternately stacked, but instead may simply be separated from each other. Thus, un-modified layer blanks 710C may be interleaved with the first and second layers if so desired. Any combination of layers can be made so long as at least one flow passage is formed in the finished flow resistor.

(237) FIG. 81 illustrates the restrictor stack 770C after finishing operations have been completed. These finishing operations can take one of two alternative forms. In the first process, the unfinished ends 771C, 772C are broken off of the restrictor stack 770C to expose the first and third cavities 761C, 763C. The exposed first and third cavities 761C, 763C form apertures 712C on first and second finished ends 773C, 774C. This results in flow passages extending from the apertures 712C on the first finished end 773C to the apertures 712C on the second finished end. Optionally, additional material removal operations can be done to the restrictor stack 770C prior to removal of the unfinished ends 771C, 772C. This has the benefit of minimizing the amount of debris which enters the flow passages, ensuring that the resulting flow restriction closely matches the theoretical flow restriction provided by the flow restrictor. Furthermore, manufacturing repeatability is greatly improved by ensuring that debris cannot enter the flow passages.

(238) In an alternative second process, the unfinished ends 771C, 772C of the restrictor stack 770C are removed through conventional material removal processes such as machining, milling turning, sawing, grinding, electrical discharge machining, or etching. Once the unfinished ends 771C, 772C are removed to form the finished ends 773C, 774C, the restrictor stack 770C is rinsed with deionized water. An electropolish process is used to dissolve any remaining metal particles and produce a surface having very low roughness. Next, deionized water is pumped through the flow passages to flush the electropolishing solution. The restrictor stack 770C is then dried and subsequently a nitric acid solution is used to remove any remaining free iron, phosphates, and sulfates. This results in a surface which is extremely clean and free of contaminants.

(239) The present invention is directed to a seal for a flow restrictor for use in an apparatus for controlling gas flow. In some embodiments, the apparatus may function as a mass flow controller to deliver a known mass flow of gas to a semiconductor or similar process. Semiconductor fabrication is one industry which demands high performance in control of gas flows. As semiconductor fabrication techniques have advanced, customers have recognized the need for flow control devices with increased accuracy and repeatability in the mass of the delivered gas flows. Modern semiconductor processes require that the mass of the gas flow is tightly controlled, the response time minimized, and the gas flow is highly accurate. The present seals ensure that the flow restrictor is sealed into its flow passage more effectively and at a reduced cost.

(240) FIG. **82** shows a schematic of an exemplary processing system **1000D** utilizing one or more flow restrictors. The processing system **1000D** may utilize a plurality of apparatus for controlling flow **100D** fluidly coupled to a processing chamber **1300D**. The plurality of apparatus for controlling flow **100D** are used to supply one or more different process gases to the processing chamber **1300D**. Articles such as semiconductors may be processed within the processing chamber **1300D**. Valves **1100D** isolate each of the apparatus for controlling flow **100D** from the processing chamber **1300D**, enabling each of the apparatus for controlling flow **100D** to be selectively connected or isolated from the processing chamber **1300D**, facilitating a wide variety of different processing steps. The processing chamber **1300D** may contain an applicator to apply process gases delivered by the plurality of apparatus for controlling flow **100D**, enabling selective or diffuse distribution of the gas supplied by the plurality of apparatus for controlling flow **100D**. In addition, the processing system **1000D** may further comprise a vacuum source **1200D** which is isolated from the processing chamber **1300D** by a valve **1100D** to enable evacuation of process gases or facilitate purging one or more of the apparatus for controlling flow **100D** to enable switching between process gases in the same apparatus for controlling flow **100D**. Optionally, the apparatus for controlling flow **100D** may be mass flow controllers, flow splitters, or any other device which controls the flow of a process gas in a processing system. Furthermore, the valves **1100D** may be integrated into the apparatus for controlling flow **100D** if so desired.

(241) Processes that may be performed in the processing system **1000D** may include wet cleaning, photolithography, ion implantation, dry etching, atomic layer etching, wet etching, plasma ashing, rapid thermal annealing, furnace annealing, thermal oxidation, chemical vapor deposition, atomic layer deposition, physical vapor deposition, molecular beam epitaxy, laser lift-off, electrochemical deposition, chemical-mechanical polishing, wafer testing, or any other process utilizing controlled volumes of a process gas.

(242) FIG. **83** shows a schematic of an exemplary mass flow controller **101D**, which is one type of apparatus for controlling flow **100D** that may be utilized in the processing system **1000D**. The mass flow controller **101D** has a gas supply of a process gas fluidly coupled to an inlet **104D**. The inlet **104D** is fluidly coupled to a proportional valve **120D** which is capable of varying the volume of process gas flowing through the proportional valve **120D**. The proportional valve **120D** meters the mass flow of process gas which passes to the P1 volume **106D**. The proportional valve **120D** is capable of providing proportional control of the process gas such that it need not be fully open or closed, but instead may have intermediate states to permit control of the mass flow rate of process gas.

(243) A P1 volume **106D** is fluidly coupled to the proportional valve **120D**, the P1 volume **106D** being the sum of all the volume within the mass flow controller **101D** between the proportional valve **120D** and a flow restrictor **160D**. A pressure transducer **130D** is fluidly coupled to the P1 volume **106D** to enable measurement of the pressure within the P1 volume **106D**. An on/off valve **150D** is located between the flow restrictor **160D** and the proportional valve **120D** and may be used to completely halt flow of the process gas out of the P1 volume **106D**. Optionally, the flow restrictor **160D** may be located between the on/off valve **150D** and the proportional valve **120D** in an alternate configuration. Finally, the flow restrictor **160D** is fluidly coupled to an outlet **110D** of

the mass flow controller **101D**. In the processing system, the outlet **110D** is fluidly coupled to a valve **1100D** or directly to the processing chamber **1300D**. In the present embodiment, the flow restrictor **160D** is located between the on/off valve **150D** and the outlet **110D**. In an alternate embodiment, the on/off valve **150D** is located between the flow restrictor **160D** and the outlet **110D**. Thus, the arrangement of the on/off valve **150D** and the flow restrictor **160D** may be reversed.

(244) Internal to the first on/off valve **150D** is a valve seat and a closure member. When the apparatus **100D** is delivering process gas, the first on/off valve **150D** is in an open state, such that the valve seat and the closure member are not in contact. This permits flow of the process gas and provides a negligible restriction to fluid flow. When the first on/off valve **150D** is in a closed state the closure member and the valve seat are biased into contact by a spring, stopping the flow of process gas through the first on/off valve **150D**.

(245) The flow restrictor **160D** is used, in combination with the proportional valve **120D**, to meter flow of the process gas. In most embodiments, the flow restrictor **160D** provides a known restriction to fluid flow. The first characterized flow restrictor **160D** may be selected to have a specific flow impedance so as to deliver a desired range of mass flow rates of a given process gas. The flow restrictor **160D** has a greater resistance to flow than the passages upstream and downstream of the flow restrictor **160D**.

(246) Optionally, the mass flow controller **101D** comprises one or more P2 pressure transducers downstream of the flow restrictor **160D** and the on/off valve **150D**. The P2 pressure transducer is used to measure the pressure differential across the flow restrictor **160D**. In some embodiments, the P2 pressure downstream of the flow restrictor **160D** may be obtained from another apparatus **100D** connected to the processing chamber, with the readings communicated to the mass flow controller **101D**.

(247) Optionally, temperature sensors may be employed to further enhance the accuracy of the mass flow controller **101D**. They may be mounted in the base of the mass flow controller **101D** near the P1 volume **106D**. Additional temperature sensors may be employed in a variety of locations, including the proportional valve **120D**, the pressure transducer **130D**, and the on/off valve **150D**.

(248) Turning to FIG. **84**, a schematic of an on/off valve **150D** is shown with a first embodiment of the flow restrictor **160D** located within an outlet passage **157D** of the on/off valve **150D**. The on/off valve **150D** has an inlet passage **158D** which allows process gas to flow into the valve **150D**. A spring **156D** biases a closure member **154D** into contact with a valve seat **152D**, preventing process gas from flowing when the valve **150D** is in a closed state. When in an open state, the closure member **154D** is moved so that it is spaced from the valve seat **152D**, allowing process gas to pass the valve seat **152D** into the outlet **157D**. The outlet **157D** is formed as a cylindrical bore, but may also be formed as an oval, polygon, or any other shape. The flow restrictor **160D** is inserted into the outlet **157D** with a seal **170D** preventing gas flow between the flow restrictor **160D** and the wall **159D** of the outlet **157D**.

(249) Turning to FIGS. **85-88**, the flow restrictor **160D** and the seal **170D** are shown in greater detail. FIG. **85** shows a perspective view of the flow restrictor **160D** and the seal **170D**. The flow restrictor **160D** extends from a first end **161D** to a second end **162D** along a longitudinal axis A-A. The seal **170D** is fitted to the flow restrictor **160D**. The seal **170D** circumferentially surrounds the flow restrictor **160D** and has an outer surface **171D**. The seal **170D** extends between a first end **172D** and a second end **173D** along a longitudinal axis B-B. The longitudinal axis B-B of the seal **170D** is collinear with the longitudinal axis A-A of the flow restrictor **160D**. However, in alternate embodiments, the longitudinal axis B-B of the seal **170D** may not be collinear with the longitudinal axis A-A of the flow restrictor **170D**. In some embodiments, the longitudinal axis B-B of the seal is angled with respect to the longitudinal axis A-A of the flow restrictor **160D**. In yet other embodiments, the longitudinal axis B-B of the seal may be spaced but parallel to the longitudinal

axis A-A of the flow restrictor **160D**. In yet other embodiments, the axes may be both angled and spaced from one another.

(250) As best seen in FIG. **86**, the flow restrictor **160D** has a sealing portion **163D** and an unsealed portion **166D**. The unsealed portion **166D** has a first diameter **D1** and the sealing portion **163D** has a second diameter **D2**, the first diameter **D1** being greater than the second diameter **D2**. The seal **170D** further comprises an inner surface **174D** which is in surface contact with the sealing portion **163D** of the flow restrictor **160D**. The outer surface **171D** has a third diameter **D3** which is greater than either of the first and second diameters **D1**, **D2**. This results in an interference fit between the wall **159D** and the outer surface **171D** and ensures that the seal **170D** seals against the wall **159D** of the outlet **157D** while simultaneously preventing contact between the flow restrictor **160D** and the wall **159D**. The inner surface **174D** defines an aperture through which the flow restrictor **160D** is received and through which all gas flows generally along the axis B-B from the first end **161D** to the second end **162D** of the flow restrictor **160D**. In yet other embodiments, the sealing portion **163D** extends the entire length of the flow restrictor **160D**. In yet further embodiments, the first diameter **D1** may be the same diameter as the second diameter **D2**. Preferably, the third diameter **D3** has an interference fit with the wall **159D**. The third diameter **D3** may also be the same diameter as the second diameter **D2**. Furthermore, the gas need not enter from the first end **161D** and exit the second end **162D** of the flow restrictor, but may also enter through the circumference of the flow restrictor **160D**. Flow of gas within the flow restrictor **160D** need not flow strictly along the axis B-B, but need only pass through the flow restrictor **160D** and past the seal **170D** rather than around it.

(251) The sealing portion **163D** has a seal receiving surface **165D** and a plurality of ridges **164D** used to improve sealing and retain the seal in place. The second diameter **D2** is reduced as compared with the first diameter **D1** so as to provide room for the seal **170D** and enhance retention of the seal **170D** on the flow restrictor **160D**. The ridges **164D** have a triangular cross-section and encircle the flow restrictor **160D**. When the seal **170D** is installed onto the sealing portion **163D** of the flow restrictor **160D**, the ridges **164D** deform the seal **170D** to further enhance the retention of the seal **170D**. This ensures that the seal **170D** is maintained on the flow restrictor **160D** when the flow restrictor is pressed into the outlet **157D**. The third diameter **D3** is typically an interference fit with the outlet **157D**, so substantial force may be required to press the seal **170D** into the outlet **157D** depending on the extent of the interference. In the exemplary embodiment, the sealing portion **163D** has two ridges **164D**. In alternate embodiments, the sealing portion **163D** may have greater or fewer ridges **164D**. The cross-sectional profile of the ridges **164D** may be rectangular, trapezoidal, or any other shape. In yet further variations, a texture may be formed on the seal receiving surface **165D**. This texture may be formed by knurling, grinding, or any other known process. In alternate embodiments, a single model of flow restrictor **160D** may be installed into a plurality of outlets **157D** having differing diameters by modifying the thickness of the seal such that the third diameter **D3** is modified to have a suitable interference with the wall **159D** of each of the respective outlets **157D**. This configuration beneficially allows the restrictor to be installed directly against the seat of the valve, greatly reducing the volume enclosed between the valve seat and the flow restrictor **160D**. In addition, multiple valve geometries, bore sizes, and fitting geometries can be accommodated by positioning the flow restrictor **160D** within the outlet **157D**.

(252) In use, process gas flows through the flow restrictor **160D** from the first end **161D** to the second end **162D**. The seal **170D** provides a close fit with both the flow restrictor **160D** and the wall **159D** of the outlet **157D** so as to prevent process gas from flowing around the flow restrictor **160D**. Although some leakage of gas is possible, this leaking is reduced to at least 1×10^{-7} atm-cc/sec when Helium is used as a process gas. This leak rate ensures that a negligible volume of process gas flows around the flow restrictor **160D** rather than through the flow restrictor **160D**.

(253) The seal **170D** is preferably formed of a non-metallic material such as a plastic material. One

exemplary material could be Polytetrafluoroethylene (also known as “PTFE” or “Teflon”). Alternate materials may include metals, ceramics, or composite materials. The seal **170D** is preferably shrunk or stretched onto the flow restrictor **160D** so as to ensure a tight fit between the seal receiving surface **165D** and the inner surface **174D**. However, other methods are contemplated. In yet further embodiments, the seal may be welded, bonded, or pressed onto the flow restrictor **160D** so as to achieve a secure gas tight connection between the seal **170D** and the flow restrictor **160D**. In yet another embodiment, a plurality of identical flow restrictors **160D** are mounted to differing seals **170D** to allow installation into different size outlets **157D**.

(254) Turning to FIGS. **89-95**, a second embodiment of a flow restrictor **260D** is shown with a seal **270D**. As can be seen in the schematic of FIG. **89**, a valve **150D** is illustrated. The valve **150D** is substantially identical to the valve **150D** of FIG. **84**. However, instead of having a flow restrictor pressed into the outlet **157D**, a seal **270D** is mounted between a sealing surface **153D** of the valve **150D** and a sealing surface **291D** of a base **290D**, the base **290D** comprising flow passages **292D** which connect the on/off valve **150D** to the various components of the mass flow controller **101D** or other apparatus for controlling flow **100D**. The seal **270D** is installed between the sealing surface **153D** and the sealing surface **291D** and has a first seal ring **271D** and a second seal ring **272D** as best shown in FIGS. **90** and **91**. The first seal ring **271D** and second seal ring **272D** are mounted to a gasket sheet **273D** and extend beyond the gasket sheet **273D** to engage sealing recesses **155D**, **295D** of the valve **150D** and the base **290D**, respectively. A plurality of apertures **274D** are provided through the gasket sheet **273D** to allow the passage of fasteners used to join the valve **150D** to the base **290D**. Additional holes **275D** may be used to facilitate manufacturing of the seal **270D** or for other purposes such as to seal additional flow passages.

(255) As can be seen in FIG. **91**, the first seal ring **271D** of the seal **270D** receives the flow restrictor **260D**. The flow restrictor **260D** extends from a first end **261D** to a second end **262D** along a longitudinal axis A-A. As best seen in FIGS. **91** and **92**, the first seal ring **271D** has a first side **276D** and a second side **277D** opposite the first side **276D**, a longitudinal axis B-B extending through the first seal ring **271D** perpendicular to the first and second sides **276D**, **277D**. The first and second sides **276D**, **277D** engage the sealing recesses **155D**, **295D** and are compressed between them when the valve **150D** is mounted to the base **290D**. The first seal ring **271D** also has an inner surface **278D** which is generally cylindrical and a sealing web **279D** which extends across the inner surface **278D**. A flow aperture **280D** is formed in the sealing web **279D** to receive the flow restrictor **260D**. The flow aperture **280D** has a generally rectangular shape in the present embodiment, but in other embodiments it may be circular, elliptical, or any other shape suitable to accommodate a corresponding flow restrictor. The flow restrictor **260D** has a generally rectangular profile along the longitudinal axis and is a close fit within the flow aperture **280D**. Once the flow restrictor **260D** is installed in the flow aperture **280D**, it can be welded, bonded, or press fit to achieve a gas tight seal between the outer surface of the flow restrictor **260D** and the sealing web **279D**, ensuring that no process gas escapes past the flow restrictor **260D** without passing through the flow restrictor **260D**. The first seal ring **271D** also has an outer surface **285D** which may be of any size or diameter so long as the first seal ring **271D** can nest within the sealing recesses **155D**, **295D**. In alternate configurations, the sealing recesses **155D**, **295D** may be omitted. In yet further configurations, the inner surface **278D** and outer surface **285D** need not be cylindrical, and may be rectangular, ellipsoid, polygonal, or any other shape.

(256) The second seal ring **272D** also has a first side **281D** and a second side **282D**. However, the second seal ring **272D** differs from the first seal ring **271D** in that it has no corresponding sealing web. Instead, the inner surface **283D** defines a flow aperture that enables the passage of process gas without significant flow impedance. Ideally, the flow passages and the second seal ring **272D** provide no restriction to fluid flow. In alternate embodiments, the seal **270D** may comprise only the first seal ring **271D** and be free of the second seal ring **272D** or any other components. Alternately, there may be more than one of the first or second seal rings **271D**, **272D**.

(257) In alternate embodiments, the flow aperture 280D of the first seal ring 271D may be circular, rectangular, have a polygon shape, may comprise arcs, or may have any known shape. Thus, any cross-section of flow restrictor may be accommodated in the seal ring 271D. In yet further embodiments, the seal ring 271D may be press fit, welded, bonded, or otherwise secured directly within a flow passage such as the outlet 157D of the valve 150D or the flow passages 292D of the base 290D. In yet further embodiments, the gasket sheet 273D maybe omitted, such that the seal is comprised only of the seal ring 271D. The seal 270D is preferably constructed at least partially of a metal material. In the most preferred embodiments, the first and second seal rings 271D, 272D are metallic.

(258) During assembly, the seal 270D is placed between the valve 150D and the base 290D and aligned so that the first and second seal rings 271D, 272D align with the sealing recesses 155D, 295D. The flow restrictor 260D then extends into the outlet 157D and the corresponding flow passage 292D in the base 290D. The flow restrictor 260D may be attached to the first seal ring 271D so that the seal is halfway along the length of the flow restrictor 260D, or it may be attached at any point along the length of the flow restrictor 260D. It may even be attached substantially flush with either the first or second end 261D, 262D. Furthermore, the seal 270D may be installed such that it is located within a portion of the valve 150D to minimize the distance between the valve seat 152D and the flow restrictor 260D, minimizing the volume therebetween. As noted previously, the seal 270D may also be configured so that the flow restrictor 260D is positioned upstream of the valve seat 152D and positioned in the inlet 158D instead of the outlet 157D. The seal of this embodiment can reliably produce a seal with a Helium leak rate better than 1×10^{-11} atm-cc/sec, substantially eliminating all flow of process gas around the flow restrictor 260D.

(259) While the invention has been described with respect to specific examples including presently preferred modes of carrying out the invention, those skilled in the art will appreciate that there are numerous variations and permutations of the above described systems and techniques. It is to be understood that other embodiments may be utilized and structural and functional modifications may be made without departing from the scope of the present invention. Thus, the spirit and scope of the invention should be construed broadly as set forth in the appended claims.

Claims

1. A method of manufacturing a semiconductor device, the method comprising: a) supplying a gas to a gas inlet of a gas flow control apparatus, the gas flow control apparatus comprising: a gas flow path extending from the gas inlet to a gas outlet; a proportional valve operably coupled to the gas flow path; an on/off valve operably coupled to the gas flow path between the proportional valve and the gas outlet, a volume of the gas flow path being defined between the proportional valve and the on/off valve; and a flow restrictor having a flow impedance operably coupled to the gas flow path and located between the proportional valve and the gas outlet; b) pressurizing the volume with the gas to a target pre-flow pressure by opening the proportional valve while the on/off valve is in an off-state, the target pre-flow pressure selected, in view of the flow impedance of the flow restrictor, to achieve the predetermined flow rate; c) opening the on/off valve by moving the on/off valve to an on-state, thereby delivering the gas to the gas outlet at the predetermined flow rate; and d) flowing the gas from the gas outlet to a semiconductor manufacturing process.
2. The method of claim 1 wherein the gas flow control apparatus further comprises a controller, the controller generating a gas flow activation signal at a first time prior to step b), the gas flow activation signal comprising data identifying a turn on time, step c) occurring at the turn on time.
3. The method according to claim 2, wherein the controller comprises an apparatus controller and a system controller, the system controller transmitting the gas flow activation signal to the apparatus controller.

4. The method according to claim 2, wherein the gas flow activation signal further comprises data identifying the predetermined flow rate.
5. The method according to claim 1, further comprising a controller, the controller generating a gas flow activation signal at a first time prior to step b), the gas flow activation signal comprising data instructing the controller to open the on/off valve upon receipt of a trigger signal, step c) occurring at a turn on time in response to the trigger signal.
6. The method according to claim 1, wherein after step b) and prior to step c), the volume is pressurized to the target pre-flow pressure and the proportional valve is closed.
7. The method according to claim 1, wherein step c) occurs as soon as the volume reaches the target pre-flow pressure.
8. The method according to claim 7, wherein the proportional valve has a flow rate substantially equal to the predetermined flow rate when step c) occurs.
9. The method according to claim 1, wherein the gas flow control apparatus further comprises a bleed valve and a bleed outlet, the bleed valve configured to vent gas from the volume to the bleed outlet when the bleed valve is in an on state and wherein the bleed valve is transitioned to an on state during step b).
10. The method according to claim 9, wherein the proportional valve has a flow rate substantially equal to the predetermined flow rate when step c) occurs.
11. The method according to claim 1, wherein step c) occurs before the volume reaches the target pre-flow pressure.
12. The method according to claim 1, wherein a priming period between step b) and step c) is constant regardless of the predetermined flow rate.
13. The method according to claim 1, wherein a priming period between step b) and step c) varies depending on the predetermined flow rate.
14. A method of manufacturing a semiconductor device, the method comprising: a) supplying a gas to a gas inlet of a gas flow control apparatus; b) generating, with a controller, a gas flow activation signal at a first time that comprises data identifying a second time at which the gas is to be delivered from a gas outlet of a gas flow path of the gas flow control apparatus at a predetermined flow rate, the first time being prior to the second time and a priming period being defined between the first and second times; c) upon receipt of the gas flow activation signal, the controller adjusting one or more components of the gas flow control apparatus to achieve a primed condition of the gas in a volume of the gas flow path during the priming period, the primed condition selected to achieve the predetermined flow rate, the gas being prohibited from exiting the gas outlet of the gas flow path during the priming period; and d) delivering the gas from the volume at the second time via the gas outlet of the gas flow path to a semiconductor manufacturing process.
15. The method according to claim 14, wherein a delivered flow rate of gas from the gas outlet stabilizes within 2% of the predetermined flow rate within 100 milliseconds after step d).
16. The method according to claim 14, wherein the controller comprises an apparatus controller and a system controller, the system controller transmitting the gas flow activation signal to the apparatus controller.
17. A method of manufacturing a semiconductor device, the method comprising: a) supplying a gas to a gas inlet of a gas flow control apparatus; b) priming, in a volume of a gas flow path and during a priming period, the gas to a primed condition, the primed condition selected to achieve the predetermined flow rate, the gas being prohibited from exiting a gas outlet of the gas flow path during the priming period; and c) delivering the gas from the volume of the gas flow control apparatus to a semiconductor manufacturing process subsequent to the priming period.
18. The method according to claim 17, wherein the volume of the gas path is defined by a proportional valve and an on/off valve, the on/off valve prohibiting the gas from exiting the gas outlet and the proportional valve priming the volume during the priming period.
19. The method according to claim 17, further comprising a controller, the controller comprising an

apparatus controller and a system controller, the system controller transmitting a gas flow activation signal to the apparatus controller at the beginning of the priming period.

20. The method according to claim 19, wherein the gas flow activation signal includes data identifying a time at which the gas is to be delivered from the gas outlet of the gas flow path at the predetermined flow rate.
